Data Centric AI

BUSINESS UNDERSTANDING & DATA UNDERSTANDING

Business Intelligence Group

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Business Understanding

Business understanding is aimed at gaining an in-depth knowledge about the domain and the business problem/opportunity:

- Involve interviews with the domain experts
- Requires the ability to quickly adapt to the context and the domain-specific vocabulary
- The most effective approach to carry out interviews is **to verify the claims made by domain experts against the data**

Data Understanding

Data understanding is aimed at gaining an in-depth knowledge about the data:

- ~~Assessing the~~ (Valutazione dei) available data
- Assessing data quality
- Carrying out a manual analysis to get initial answers or to form hypotheses about the problem
- Verify the claims made by the domain experts and develop further questions to ask them

The type of data determines the methods (the algorithm that we can apply),
but data quality determines performance

Types of data

Different types of data imply different expressiveness and different techniques to be adopted to analyze them

- Tabular (structured)
- Hierarchical
- Graphs
- Text (NLP)
- Images
- Audio

Structured Data

- Structured data are composed of collections of *objects* described by a set of attributes
  - ✓ Synonyms for object are record, point, case, example, entity, instance, element
- An *attribute* is a property or characteristic of an object
  - ✓ Synonyms for attribute are: variable, field, characteristic

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

è fondamentale comprendere la natura dell'attributo che si sta analizzando: le sue proprietà determinano quali operazioni matematiche e statistiche abbiano senso e quali no.

Attribute Types

In order to perform meaningful analysis the characteristics of the attributes must be known

- An employee is described by an ID and age, but it makes no sense to calculate the average employee ID! ID is unique, so it cannot be generalised
- The attribute type tells us what properties of the attribute are reflected in the value we use as a measure
- A simple way to characterize different types of attributes is based on the type of operator it makes sense to apply to the values it takes:
  - ✓ Diversity =, ≠ Attributi Nominali hanno solo operatori di diversità
  - ✓ Ordering <, ≤, >, ≥ Attributi Ordinali hanno solo operatori di diversità e ordinamento
  - ✓ Additivity +, - Attributi ad intervallo hanno solo gli operatori di diversità, ordinamento e Additività
  - ✓ Multiplicativeness *, / Attributi a Rapporto hanno tutti gli operatori
- This determines 4 types of attributes: nominal, ordinal, interval, and ratio

Attribute Types

Type		Description	Example	Statistical operator
Categorical (qualitative)	Nominal	Different names of values. We can only distinguish one from the others (no sorting capabilities)	Gender, eye color, zip code, ID	Mode, correlation
	Ordinal	Values enable us to sort objects based on the value of the attribute, but the distance between consecutive values is not measurable or necessarily equal	Grade, Hardness of a mineral	Median, percentile
Numerical (quantitative)	Interval	The difference between the values has a meaning, i.e., there is a unit of measurement (i.e. and equal numerical intervals) (no ratio capability: non esiste uno zero assoluto)	Dates, temperature in Celsius and Fahrenheit	Mean, variance
	Ratio	Significato: Esiste uno zero assoluto e non arbitrario che rappresenta la totale assenza della grandezza misurata. Perché il rapporto ha senso: 0 kg significa assenza di peso. Di conseguenza, un oggetto di 20 kg pesa esattamente il doppio di uno di 10 kg. ↑ The ratio of values has meaning	Age, mass, length, amount of money, temperature expressed in Kelvin	Geometric mean, harmonic mean

Temperature in Celsius
- - 10°C → 0°C → 10°C → 20°C → 30°C
- The difference makes sense: 30°C − 10°C = 20°C
- **The ratio doesn't make sense**: 20°C is not "the double" of 10°C, because 0°C doesn't mean "Absence of heat".
- How would you describe -10° C with respect to 20° C in terms of ratio?

				…cal operator
				…correlation
…e)		Ordinal	attribute, but the distance between consecutive … not measurable or necessarily equal	…eral	Median, percentile
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		Ratio	The ratio of values has meaning	Age, mass, length, amount of money, temperature expressed in Kelvin	Geometric mean, harmonic mean

Weight in kg

- 0 kg → 10 kg → 20 kg → 30 kg
- The difference makes sense: 30 kg − 10 kg = 20 kg
- The ratio makes sense: 20 kg is twice 10 kg since 0 kg means no weight

			Example	Statistical operator
			Gender, eye color, zip code, ID	Mode, correlation
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Attribute Types: Alternative Classifications

- Binary, discrete and continuous
  - ✓ A discrete attribute has a finite number or an infinite countable set of values normally represented by integers or labels
  - ✓ A continuous attribute takes ~~real values~~ valori reali (insieme R)
  - ✓ Nominal and ordinal attributes are typically discrete or binary, while interval and ratio attributes are continuous
- Asymmetric attributes: only instances that take non-zero values are relevant
  - ✓ Consider student records: where each attribute represents a course at the University that may be taken (1) or not (0) by the student. Since students take a very small fraction of the courses in the Athenaeum if one were to compare students' choices based on all attribute values their behavior would appear very similar.

Attribute Types: Alternative Classifications

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An infinite set is said to be *countably infinite* if its elements can be enumerated by the natural numbers; equivalently, there exists a bijective mapping between the set and $\mathbb{N}$

- Positive
- Integers
- Even/Odd numbers
- The Cartesian product of two countably infinite sets
- Rational numbers
- All finite strings over a finite alphabet

Examples

…relevant

… the University that may be taken (1) …e courses in the Athenaeum if one …behavior would appear very similar.

Un attributo è definito asimmetrico quando solo la presenza del valore (valore non-zero o 1) fornisce informazione rilevante, mentre la comune assenza (valore 0) non implica alcuna vera similarità tra gli oggetti.

Attribute Types: Asymmetric Attributes

- **Asymmetric attributes**: only the presence of a value (non-zero) is relevant; sharing a 0 (a common absence) does not make two objects similar
  - ✓ Study plan: a binary vector with one attribute for each exam offered by the University (1 = chosen, 0 = not chosen)
  - ✓ The University offers m = 1,000 exams while each student chooses about 20 of them: more than 98% of the values are 0

L'attributo asimmetrico è il singolo esame universitario. L'oggetto è lo studente dove ciascuna riga della tabella corrisponde a uno specifico studente descritto dal vettore delle sue scelte sui 1000 esami disponibili.

- The similarity between study plans must be based on the **common exams** only
  - ✓ f_{ab} = number of exams with value a in the first plan and b in the second one. The Simple Matching Coefficient (SMC) counts both 1–1 and 0–0 matches, the Jaccard coefficient (J) ignores 0–0 matches

f11 indica quanti esami hanno entrambi gli studenti
f01 o f10 indica quanti esami di uno studente differiscono dall'altro studente
f00 indica quanti esami uno studente non ha scelto

$$SMC = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}} \qquad J = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}$$

0	1	0	0	0	0	1	…	0	0	0	1	0	1	0	0
0	1	0	0	0	0	0	…	0	0	0	1	0	0	0	1

Anna and Marco choose 20 exams each, 2 in common:

- $f_{11} = 2$, $f_{10} = f_{01} = 18$, $f_{00} = 962$
- SMC = 964 / 1000 = 0.96 → **almost identical!**
- J = 2 / 38 = 0.05 → **very different**

Attribute Types: Documents & Texts

Non ci focalizziamo su questo durante il corso per quanto riguarda tipi di attributi

- Documents are the objects of the analysis; they are described by a vector of terms
  - ✓ Each term is an attribute of the document
  - ✓ The value of the attributes indicates the number of times the corresponding term appears in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Attribute Types: Transactions

- A record where:
  - ✓ Each record (transaction) involves multiple items
  - ✓ Items come from a finite set
  - ✓ For example, in a supermarket, the set of products purchased by a customer during a visit to the store constitutes a transaction, while the set of items is the set of all the products sold by the supermarket
  - ✓ The number of items may vary from transaction to transaction

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Attribute Types: Ordinal and Time series

Both involve an order, but there's a fundamental difference related to the meaning of the order and time

- Ordered data: Data organized according to a specific order, but the order does not necessarily represent time. You can compare positions or rank data, but there is no implied time interval
  - Student rankings by score (1st, 2nd, 3rd)
  - Clothing sizes (S, M, L, XL)
  - Satisfaction levels (1–5 stars)
- Time series data: Data collected in ~~temporal sequence~~ sequenza temporale, where each value has a precise timestamp often with regular or variable frequency. Time is intrinsic; patterns, trends, and correlations depend on the temporal sequence.
  - Daily temperature
  - Stock prices minute by minute
  - Hourly energy consumption

L'ordine presente nei dati non ha alcuna relazione con il tempo. La sequenza definisce solo una gerarchia, una precedenza o una posizione relativa (cosa viene prima e cosa viene dopo), ma non indica quando i valori si manifestano o si registrano.

Attribute Types: Ordinal and Time series

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- Ordered data: Data organized according to a specific order, but the order does not necessarily represent [illegible] compare positions or rank data, but there is no implied time interval

Ordinal data are the standard in

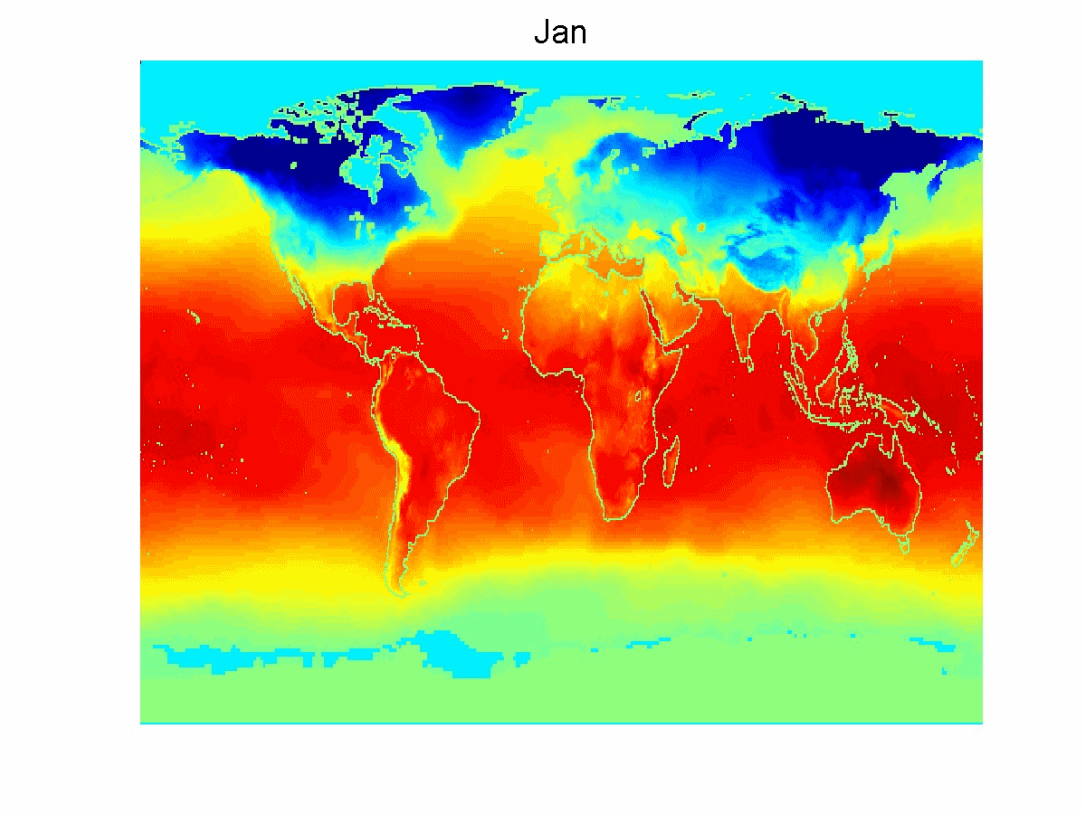
1. **Market research and user experience** In surveys like "rank your preferences," users are asked to order products, services, or features from most to least important.
2. **Psychology and cognitive sciences** Preference tests or cognitive games in which participants must rank stimuli (images, words, scenarios) according to liking, emotional response, or perceived difficulty. The ranking becomes research data to understand perceptions and mental patterns.
3. **Medicine and health** In some clinical studies, patients rank symptoms or side effects by severity or impact on quality of life. The ranking helps researchers assess subjective priorities without requiring precise numerical scales.
4. **Recommendation systems** In Netflix, Spotify, or Amazon, users can rank or prioritize movies, songs, or products they prefer. The algorithm uses this information to personalize suggestions, even without explicit numerical ratings.

Ordering becomes an indirect method of data collection, because **it allows subjective perceptions, preferences, or relative priorities to be transformed into structured information without requiring absolute numbers**.

Attribute Types: Spatio-Temporal

I dati spazio-temporali rappresentano fenomeni misurati contestualmente nello spazio (coordinate geografiche o posizioni relative) e nel tempo (timestamp o intervalli cronologici). La loro caratteristica fondamentale è la dipendenza dai "vicini" (neighbors), la quale fa sì che il valore di un punto sia strettamente legato e condizionato dai valori dei punti adiacenti sia dal punto di vista geografico sia da quello temporale.

- In Spatio-Temporal Data, values are related to the “neighbors” data both in space and time



Univariate Analysis

Univariate and Bivariate can be done manually (statistically); Multivariate instead must be done by algorithms

/feature on its own

Univariate analysis refers to the examination and understanding of a single variable in a dataset. Its purpose is to explore the distribution, central tendency, spread, and potential anomalies of that variable before using it in a model.

- Statistical Summary: Basic statistics like mean, median, mode, standard deviation, min/max are calculated to summarize the variable's behavior.
- Feature Insights: By analyzing one variable at a time, you can understand its range, skewness, and variability, which informs feature engineering and preprocessing steps like normalization or encoding.
- Visualization: Techniques include histograms, boxplots, bar charts, and density plots to visually inspect patterns or irregularities.
- Understanding Data Quality: It helps identify missing values, outliers, or inconsistent entries, which are critical in data-centric approaches where the quality of data often matters more than the complexity of the model.

Statistical Summary for Categorical Attributes

The main statistics to collect are:

- Frequency Counts – Number of occurrences of each category
  - ✓ Give the attribute 'Municipality of residence' for the Italian Citizen Dataset, the 'Bologna' value appears in about 0.6% of cases (~3.7×10^5 / 6×10^7).
- Relative Frequency / Proportions – Percentage of each category relative to the total
  - Example: 50% Red, 30% Blue, 20% Green
- Mode – The most frequent category
  - ✓ The mode value for the attribute 'Municipality of residence' for the Italian Citizen Dataset is 'Rome' that appears in about 4.5% of cases (~2.7×10^6 / 6×10^7)
- Unique Values / Cardinality – Number of distinct categories.
  - Helps identify high-cardinality variables that may need special handling.
- Missing Values – Count and proportion of missing or null entries.

Statistical Summary for Numerical Attributes

The main statistics to collect are:

Most important statistics

- **Central Tendency** — describes where is the central point (individuano il punto o il valore attorno al quale i dati tendono a raggrupparsi)
  - Mean – Average value
  - Median – Middle value (robust to outlier)
  - Mode – Most frequent value
- **Dispersion** – describes **how spread out the values are**
  - Standard Deviation – Average spread around the mean
  - Variance – Squared dispersion
  - Range – Difference between max and min
  - Interquartile Range (IQR) – Spread between Q1 and Q3 (robust to outliers)
- **Distribution Shape** — Questi indici descrivono la forma della distribuzione rispetto a una distribuzione teorica simmetrica o normale
  - Skewness – Measures asymmetry (left/right tail)
  - Kurtosis – Measures “tailedness” or peakness of the distribution

La dispersione indica quanto i dati siano "sparpagliati" attorno a un punto centrale di riferimento (solitamente la media), quindi quanto dista un certo punto dal punto centrale.

Forniscono punti di riferimento per comprendere dove si collocano specifici sottoinsiemi di valori lungo la distribuzione

- **Position** – Describes where values lie within the distribution.
  - Minimum / Maximum
  - Quartiles (Q1, Q2/median, Q3)
  - Percentiles (e.g., 5th, 95th)
- **Missing Values** – Count and proportion of missing or null entries.

La mediana è robusta agli outlier (valori anomali o estremi) grazie alla sua natura posizionale anziché aritmetica:
- Calcolo basato sull'ordinamento e non sul valore esatto: Per calcolare la mediana, i dati vengono disposti in ordine crescente. La mediana identifica semplicemente l'elemento al centro della distribuzione.
- Insensibilità all'entità degli estremi: Se il valore più grande di un dataset viene sostituito con un numero immensamente più alto, la posizione dell'elemento centrale non subisce alcuna variazione. L'outlier, indipendentemente dalla sua grandezza numerica, pesa sempre e soltanto come una singola osservazione situata all'estremità della graduatoria.

Measures of position: mean and median

- The **mean** is the most common measure for locating a set of points

$$mean(\mathbf{x}) = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Unfortunately, the mean is very sensitive to outliers

La media include direttamente tutti i valori nella sua somma. Di conseguenza, un solo dato estremamente alto o basso altera significativamente il totale della sommatoria, trascinando e distorcendo l'intera media verso l'outlier.

- In many cases it is preferred to use the median or a "controlled" mean. (Median is robust to outliers)
- In a set n of data ~~arranged~~ sorted in ascending order, the **median** is the term occupying the central place if the terms are odd; if the terms are even, the median is the arithmetic mean of the 2 central terms.

$$median(\mathbf{x}) = \begin{cases} x_{m+1} & if\ n \text{ is odd } n = 2m+1 \\ (x_m + x_{m+1})/2 & if\ n \text{ is even } n = 2m \end{cases}$$

La varianza misura il grado di dispersione dei dati elevando al quadrato la distanza di ciascuna osservazione dalla media aritmetica (Più alto è il valore della varianza, maggiore è la dispersione complessiva del dataset). La deviazione standard indica lo scostamento o la distanza media dei valori rispetto alla media aritmetica (Una deviazione standard bassa significa che i dati sono strettamente raggruppati attorno alla media. Una deviazione standard elevata significa che i punti sono ampiamente distribuiti lungo la scala).

Measures of dispersion: Range and Variance

- Range is the difference between the minimum and maximum values taken by the attribute
- Variance and standard deviation are the most common measures of dispersion of a data set.

$$Variance(\mathbf{x}) = s_{\mathbf{x}}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad StandDev(\mathbf{x}) = s_{\mathbf{x}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- Variance is sensitive to outliers since it is quadratically related to the concept of mean
- Other measures less sensitive to this problem are:

outliers

MedianAbsDev $MAD(\mathbf{x}) = median(\{|x_1 - \bar{x}|, \ldots, |x_n - \bar{x}|\})$

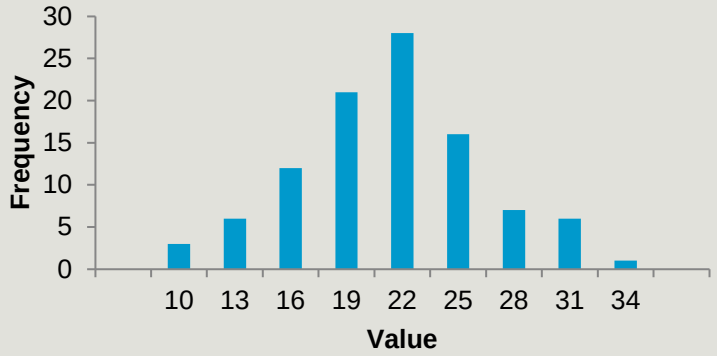
Calcola la mediana dei valori assoluti degli scarti delle osservazioni rispetto alla media.

Saltato ⟷ AbsAvgDeviation $AAD(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}|x_1 - \bar{x}|$

InterquartileRange $RI(\mathbf{x}) = x_{75\%} - x_{25\%}$

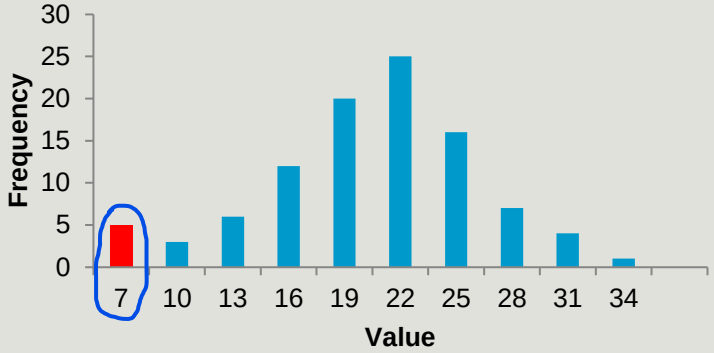
Rappresenta la distanza tra il 75esimo percentile ed il 25esimo percentile della distribuzione.

Measures of dispersion: Range and Variance



Mean 19,82617
Median 19,65625
25% quartile 16,79252
75% quartile 22,75032

Variance 25,31324
StandDev 5,031227
RI 5,957806
AAD 3,857429
MAD 2,979841



Mean 18,67617
Median 19,27243
25% quartile 15,25606
75% quartile 22,55218

Variance 37,58087
StandDev 6,130324
RI 7,29612
AAD 4,579804
MAD 3,095489

Misure di posizione:

Percentiles and Quartiles

un attributo ordinale o continuo

Given ~~an ordinal or continuous attribute~~ x and a number p between 0 and 100, the p-th percentile is the value of x_p of x such that p% of the observed values for x are lower than x_p.

- For the attribute "height in centimeters" for the population of Italian female infants at one year of age is:
  - 50th percentile= 78 cm (x_p) -> half of the girls are taller than 78 cm
  - 97th percentile= 81 cm (x_p) -> only 3% of girls are taller than 81 cm

Al 50esimo percentile della distribuzione, solo il 50% è più alta di 78cm
Al 97esimo percentile della distribuzione, solo il 3% è più alta di 81cm

- Percentiles are useful when high precision is needed, for example to define thresholds, normalize data, or analyze distribution tails (e.g., 95th or 99th percentile).

Quartiles are a special case of percentiles: they divide the data into 4 equal parts.

- Q1 (first quartile) = 25th percentile
- Q2 (second quartile) = 50th percentile (the median)
- Q3 (third quartile) = 75th percentile
- Quartiles are often used for quick analysis and to compute the IQR (InterQuartileRange) (useful for detecting outliers).

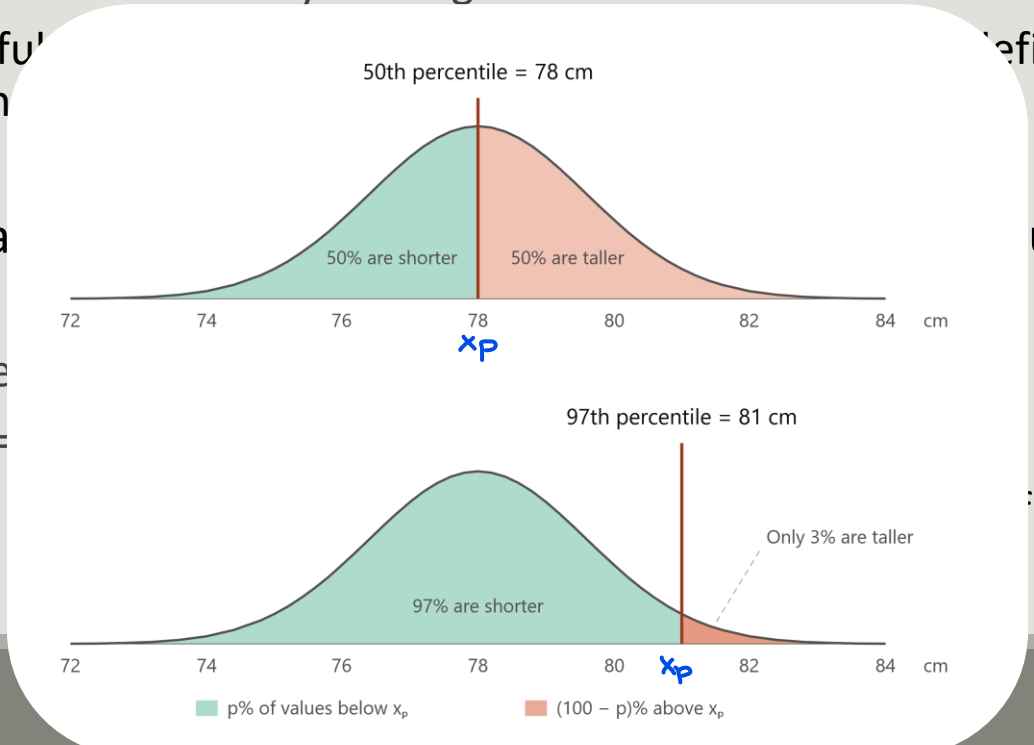
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Quartiles are a special ca… ual parts.

- Q1 (first quartile) =…
- Q2 (second quartile…
- Q3 (third quartile) =…
- Quartiles are often … ul for detecting outliers).



Skewness

La Skewness è una misura di forma

quanto la distribuzione dei dati sia asimmetrica rispetto alla sua media aritmetica.

Skewness measures ~~how asymmetrical a distribution is relative to the mean.~~

$$Skew(\mathbf{x}) = \gamma(\mathbf{x}) = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^3}{\sigma^3} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^3}{(\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2})^3}$$

$\gamma(\mathbf{x}) = 0$ Symmetric distribution
$\gamma(\mathbf{x}) > 0$ Positive asymmetry (longer right tail)
$\gamma(\mathbf{x}) < 0$ Negative asymmetry (longer left tail)

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Why do we raise to the third power?

- If we just compute $x_i - \bar{x}$ we would get 0 by definition of mean
- Squaring $(x_i - \bar{x})^2$ gives the variance but loose the sign
- Cubing $(x_i - \bar{x})^3$ preserve the sign
  - Positive deviations (values above the mean) contribute positively.
  - Negative deviations (values below the mean) contribute negatively.

Di conseguenza, la coda dominante imporrà il proprio segno al risultato finale della sommatoria.

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$\gamma(\mathbf{x})$ = 0 Symmetric distribution

$\gamma(\mathbf{x})$ > 0 Positive asymmetry (longer right t

We divide by the cubing of the standard deviation σ^3 to:

- Make skewness **dimensionless** (so it doesn't depend on the units of measurement).
- Standardize it so that we can **compare distributions of different scales**.

il risultato ottenuto non ha un'unità di misura

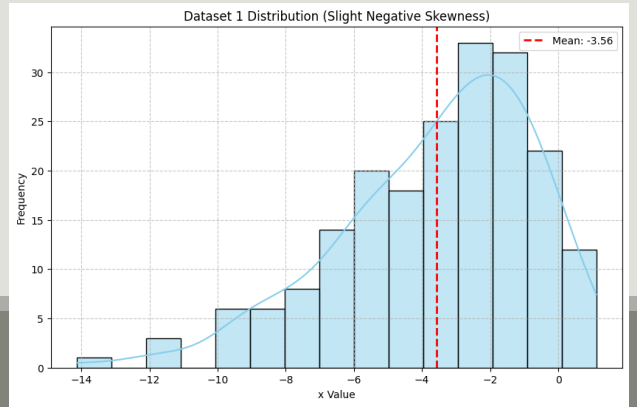
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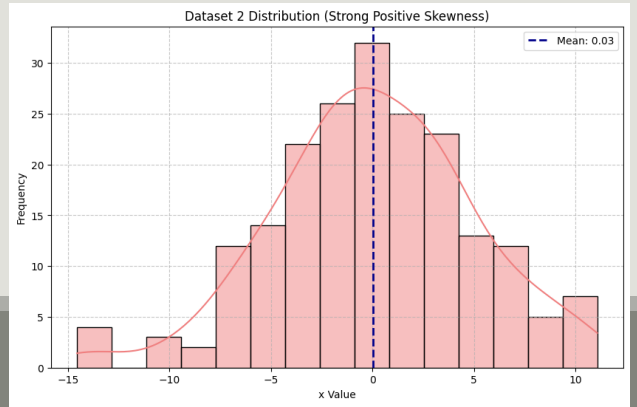
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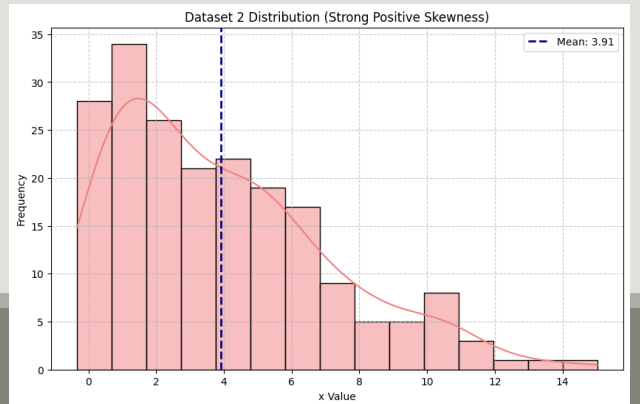
Skewness: -0.8226



Skewness: -0.2071



Skewness: 0.9127



Syntetic_Skewness.ipynb

Kurtosis

La Kurtosis misura la "pesantezza delle code" di una distribuzione, ovvero la propensione di una variabile a generare valori estremamente lontani dalla media (outlier).

~~Kurtosis measures how heavy the tails of a distribution are (propensity to extreme values)~~.

$$Kurt(\mathbf{x}) = \kappa(\mathbf{x}) = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{\sigma^4} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{(\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2})^4}$$

$\kappa(\mathbf{x}) = 3$ Mesokurtic (tails as heavy as in the normal distribution)
$\kappa(\mathbf{x}) > 3$ Leptokurtic (heavier tails: more extreme values)
$\kappa(\mathbf{x}) < 3$ Platykurtic (lighter tails: fewer extreme values)

Excess kurtosis = $\kappa(\mathbf{x}) - 3$ (= 0 for the normal distribution)

Kurtosis

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$\kappa(\mathbf{x})$ = 3 Mesokurtic (tails as heavy a

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$\kappa(\mathbf{x})$ < 3 Platykurtic (lighter tails: fe

Excess kurtosis = $\kappa(\mathbf{x})$ − 3 (= 0 for th

Why do we raise to the fourth power?

- As in the variance, an even power makes all deviations positive: the **direction** of a deviation is lost (it is measured by skewness)
- Compared with the square, the fourth power **amplifies large deviations**: a value at 1σ from the mean contributes 1, at 2σ contributes 16, at 3σ contributes 81 (16=2^4; 81=3^4)
- Kurtosis is therefore dominated by the few values in the **tails**, not by the values close to the mean

Kurtosis

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$\kappa(\mathbf{x})$ = 3 Mesokurtic (tails as heavy as in the norm

We divide by σ^4 to make kurtosis **dimensionless** and to compare distributions of different scales. Since κ = 3 for every normal distribution, tools often return the **excess kurtosis** κ − 3 (e.g., pandas .kurt(), scipy.stats.kurtosis)

What does kurtosis add with respect to skewness?

- Skewness tells **on which side** of the mean the deviations are; kurtosis tells **how extreme** the largest deviations are, whatever their side
- Distributions with the same mean, variance and skewness can still differ: values spread evenly vs. many values close to the mean plus a few very far away (potential outliers)

Kurtosis

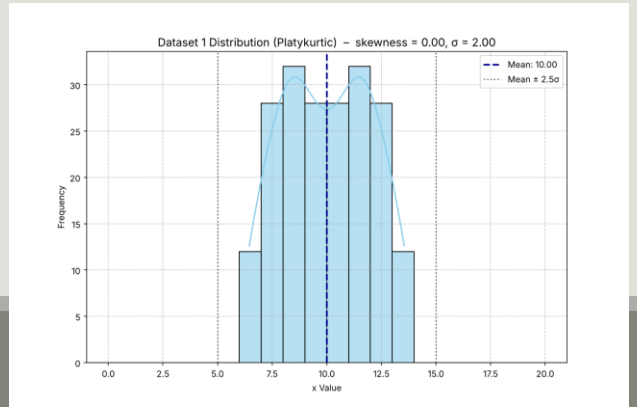
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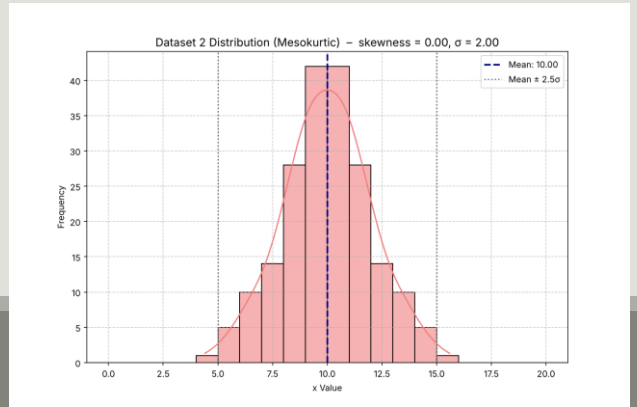
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Excess kurtosis = $\kappa(\mathbf{x})$ − 3 (= 0 for the normal distribution)

Same mean (10), standard deviation (2) and skewness (0): **only kurtosis tells the three datasets apart.**
Dataset 3 has more values close to the mean and more values beyond ±2.5σ (4% vs. 0% and 1%).

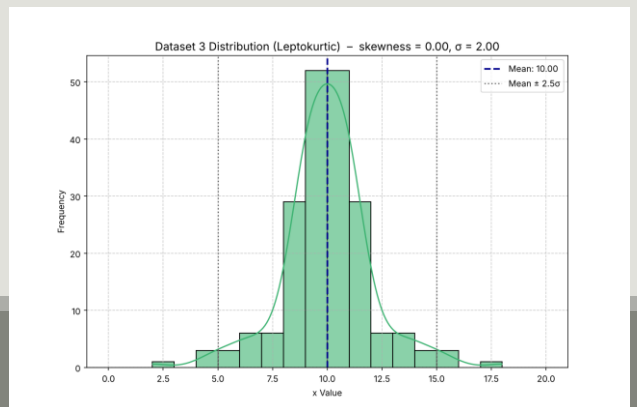
Kurtosis: 1.8631



Kurtosis: 2.9910



Kurtosis: 5.8157



Synthetic_Kurtosis.ipynb

Rappresenta cosa definisce esattamente il singolo record del dataset (es. una singola transazione al secondo, il totale degli acquisti giornalieri di un cliente o il fatturato mensile di un negozio).

Nella fase di Data Understanding, il profiling dei dati si estende oltre le sole statistiche descrittive per valutare tre aspetti strutturali fondamentali del dataset: la granularità, la qualità e la disponibilità delle etichette.

Additional Profiling Information

Other information to collect when you are analysing the data

Data Granularity: *which is the level of details of raw data?*

Example: get the temperature every three minutes is more precise than every 3 hours (need to consider if it carries more information or noise)

- Setting the right level of abstraction of data provided to a model impact on performances:
- Can hide or emphasize patterns, bias

Scegliere il corretto livello di astrazione è determinante per l'efficacia degli algoritmi.
- Troppo dettaglio: Può introdurre rumore irrilevante e far apprendere al modello variazioni casuali anziché la tendenza di fondo.
- Troppa aggregazione: Può nascondere o enfatizzare pattern e bias nascosti. Ad esempio, aggregando i dati di vendita su base mensile si potrebbero perdere i picchi di acquisto durante i fine settimana o mascherare anomalie locali legate a specifiche fasce orarie.

Data quality: *which errors are present in the data?*

- missing values
- noise
- inconsistencies between data: contrasti logici tra attributi o discrepanze di formato (es. un'età negativa, una data di nascita posteriore a un titolo di studio, o la stessa categoria scritta con sintassi diverse).
- duplicates
- measurement errors

Label Availability: *Are labels available? How many labeled instances exist? Are labels reliable?*

We will discuss this points in the Data Preparation step

Bank Data

bank-data.csv dataset

- ✓ 600 istances of Bank customers
- ✓ No missing data

Statistical Summary – Univiarate.ipjnb

A Personal Equity Plan (PEP) was a tax-advantaged investment account promoted by the United Kingdom Government to (1) Encourage Household Saving; (2) Promote Long-Term Investment; (3) Support Capital Markets

Attribute	Description
Id	Unique identifier
Age	Customer age in years (numeric)
Sex	MALE / FEMALE
Region	inner_city/rural/suburban/town
Income	Customer income (numeric)
children	Number of children? (numeric)
car	Does she own a car? (YES/NO)
save_acct	Does she have a save account? (YES/NO)
current_acct	Does she have a current account? (YES/NO)
Mortgage	Does she subsribed a mortgage?(YES/NO)
Pep	**Have she purchased a Personal Equity Plan (PEP) since the last mailing? (YES/NO)** Need this dataset for Classification Task

Bivariate Analysis

Bivariate analysis is the study of the relationship between two variables. It helps reveal dependencies, correlations, or patterns that can be critical for feature engineering, data cleaning, and model design.

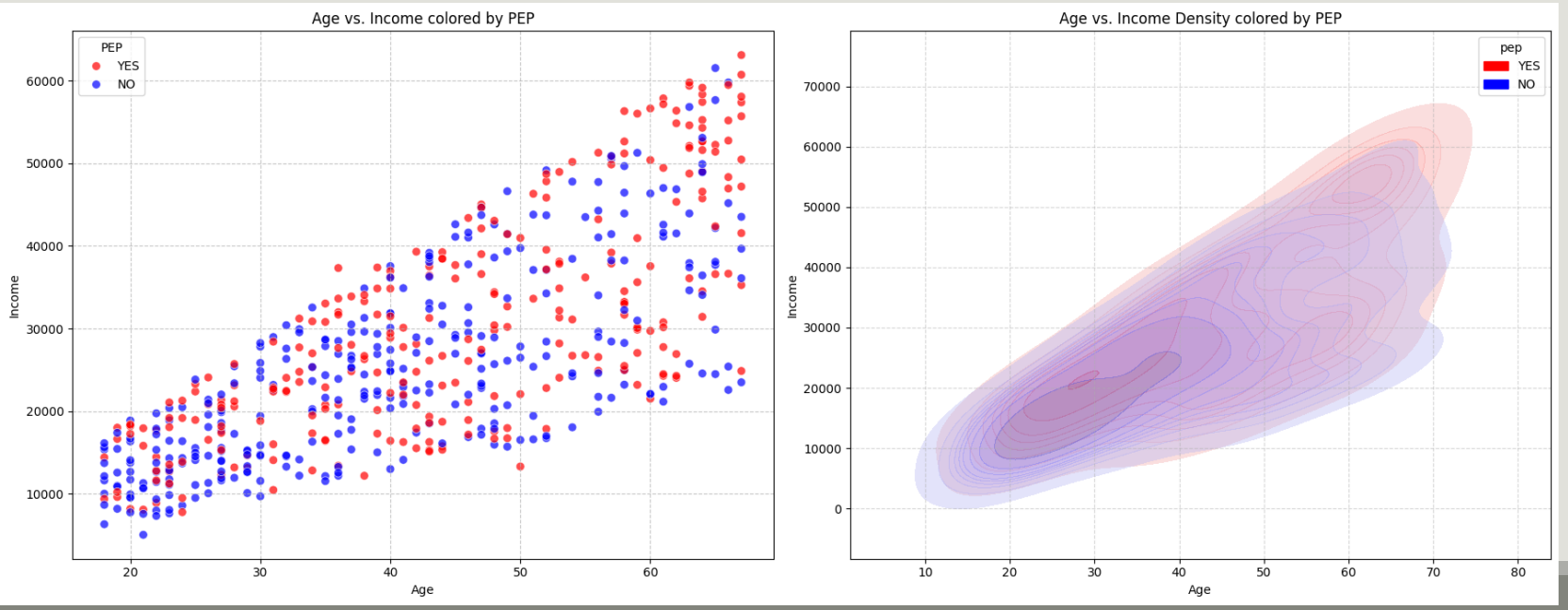
- Visual inspection: for all the attribute types (see Statistical_Summary – Biviarate.ipjnb)
- Correlation Assessment: for numerical variables, it often involves calculating correlation coefficients (like Pearson, Spearman) to quantify the strength and direction of relationships.
  - Linear correlation: Pearson and Spearman correlation
  - Non-Linear correlation: Mutual Information

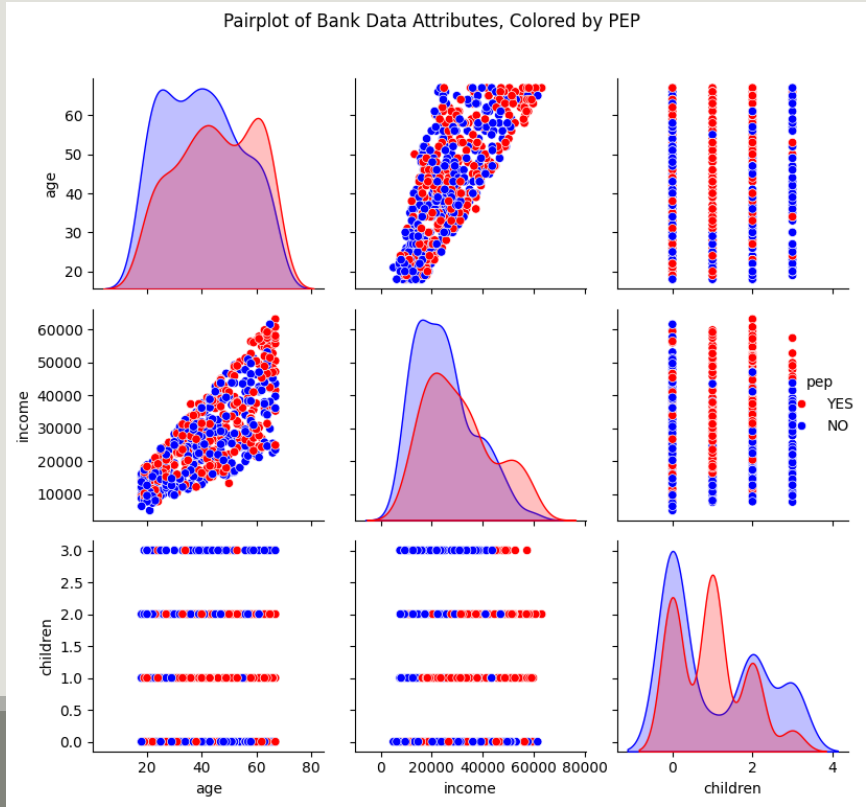
Bivariate Analysis: Visual Inspection

The best visualization depends on the involved variable types and dataset cardinality

- Continuos vs Continuos
  - Scatter plots when the number of points is manageable (raw data insight)
  - Heatmap when the number of points is large and you need aggregation

Positive Linear Correlation





Bivariate Analysis: Visual Inspection

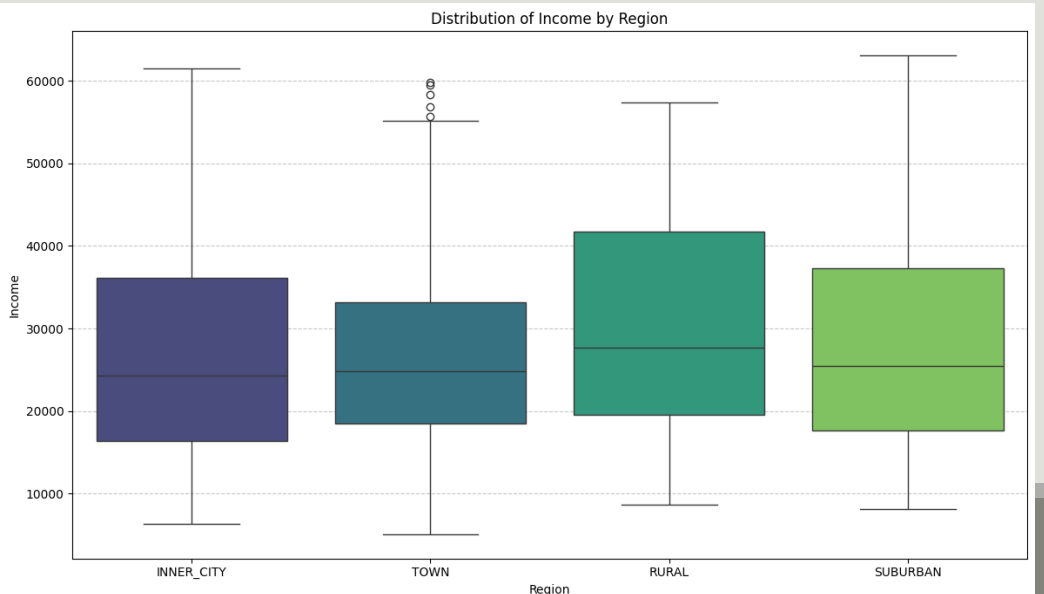
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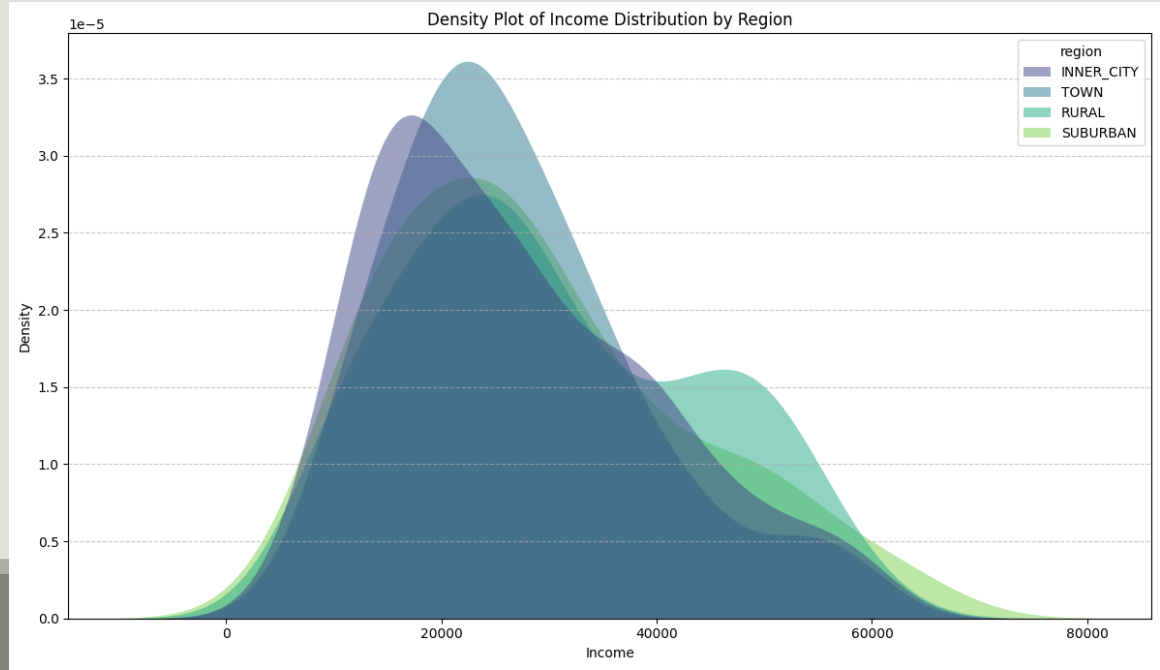
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- Categorical vs Continuos
  - Box plot: one for each value of the categorical attribute
  - Density plot

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Bivariate Analysis: Visual Inspection

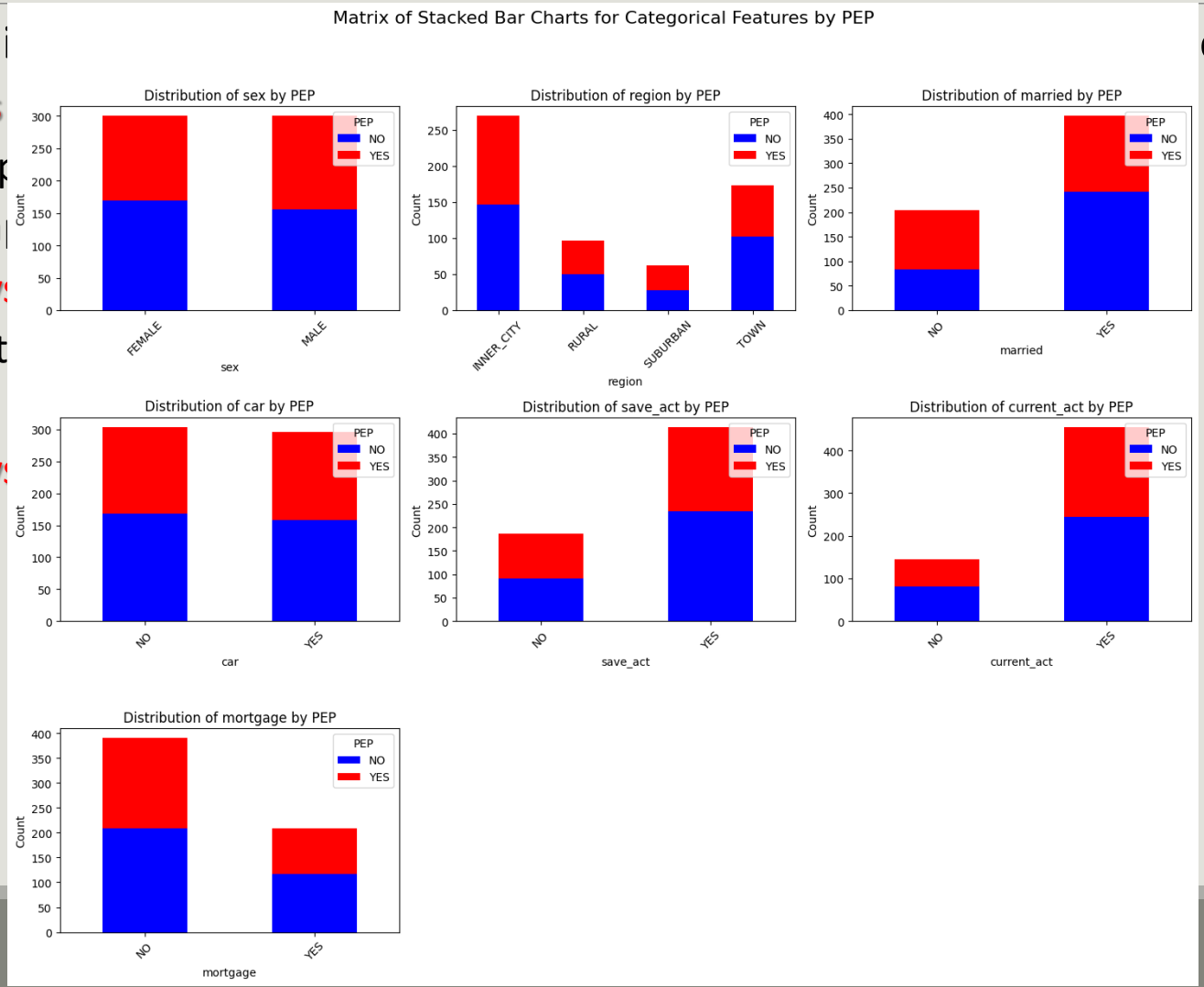
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- Categorical vs Continuos
  - Box plot: one for each value of the categorical attribute
  - Density plot
- Categorical vs Categorical
  - Stacked Bar Chart: X-axis - v_1, Color - v_2, Height - counts or proportions

Bivariate Analysis: Visual Inspection

The best visuali cardinality

- Continuos vs
  - Scatter p
  - Heatma
- Categorical v
  - Box plot
  - Density
- Categorical v
  - Stacked



Bivariate Analysis: Dataset Slicing

Data profile can differ for different portions of the dataset that should be analyzed by subpopolation

- Geographic regions
- Demographic categories
- Operating conditions

Pearson Correlation

- The correlation between pairs of objects described by attributes (binary or continuous) is a measure of the existence of a linear relationship between its attributes

$$Corr(\mathbf{x},\mathbf{y}) = \frac{Cov(\mathbf{x},\mathbf{y})}{StDev(\mathbf{x}) \cdot StDev(\mathbf{y})}$$

Valutare la modifica di una variabile rispetto ad un altra. Da questo calcolo, possiamo valutare correlazione positiva e negativa

$$Cov(\mathbf{x},\mathbf{y}) = \frac{1}{n}\sum_{k=1}^{n}(x_k - \bar{x})(y_k - \bar{y})$$

$$StDev(\mathbf{x}) = \sqrt{\frac{1}{n}\sum_{k=1}^{n}(x_k - \bar{x})^2}$$

- Pearson correlation normalizes covariance by the product of standard deviations → makes it scale-free
- The correlation varies between [-1,1]. = 0 se le due variabili non sono correlate
  - ✓ A correlation of 1 (or -1) means that attributes can be mutually expressed by a linear relationship of the type $x = a\,y + b$ We can find a linear function that perfectly describe the relation

Pearson Correlation

x=(-3, 6, 0, 3, -6) **y**=(1,-2, 0, -1, 2) Corr(**x**,**y**)=-1

x=(3, 6, 0, 3, 6) **y**=(1,2, 0, 1, 2) Corr(**x**,**y**)=1

- There could still be non-linear relationships between the data that would therefore not be captured!
  by linear correlation like Pearson
  - ✓ Between the following objects there is a correlation of the type $x_k=y_k^2$ but Corr(***x***,***y***)=0

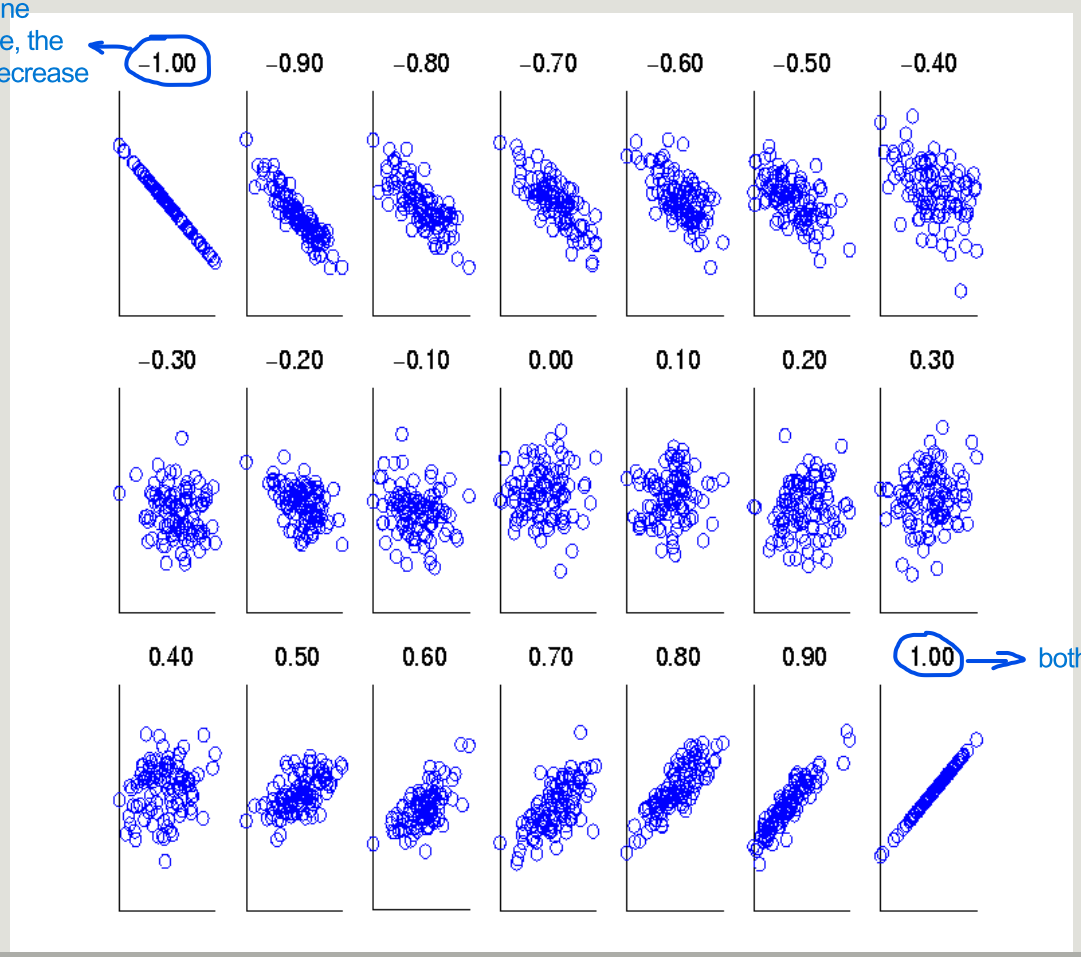
    x=(-3, -2, -1, 0, 1, 2, 3) ***y***=(9, 4, 1, 0, 1, 4, 9)

- Correlation can also be useful for discarding attributes that do not carry additional information
  - ✓ In this case x and y represent two distinct attributes and their elements the instances of the two attributes in the different objects of the data set

Quando due feature mostrano un'elevata correlazione, una delle due può essere eliminata in quanto ridondante: l'informazione portata dalla seconda è già implicitamente contenuta (e ricavabile) nella prima attraverso una relazione lineare del tipo x = ay + b. Quindi, è possibile scartarne una per ridurre la dimensionalità del dataset senza perdita di informazione, essendo il suo comportamento espresso in modo deterministico dall'æa a.

Visualizing Pearson Correlation

when one increase, the other decrease



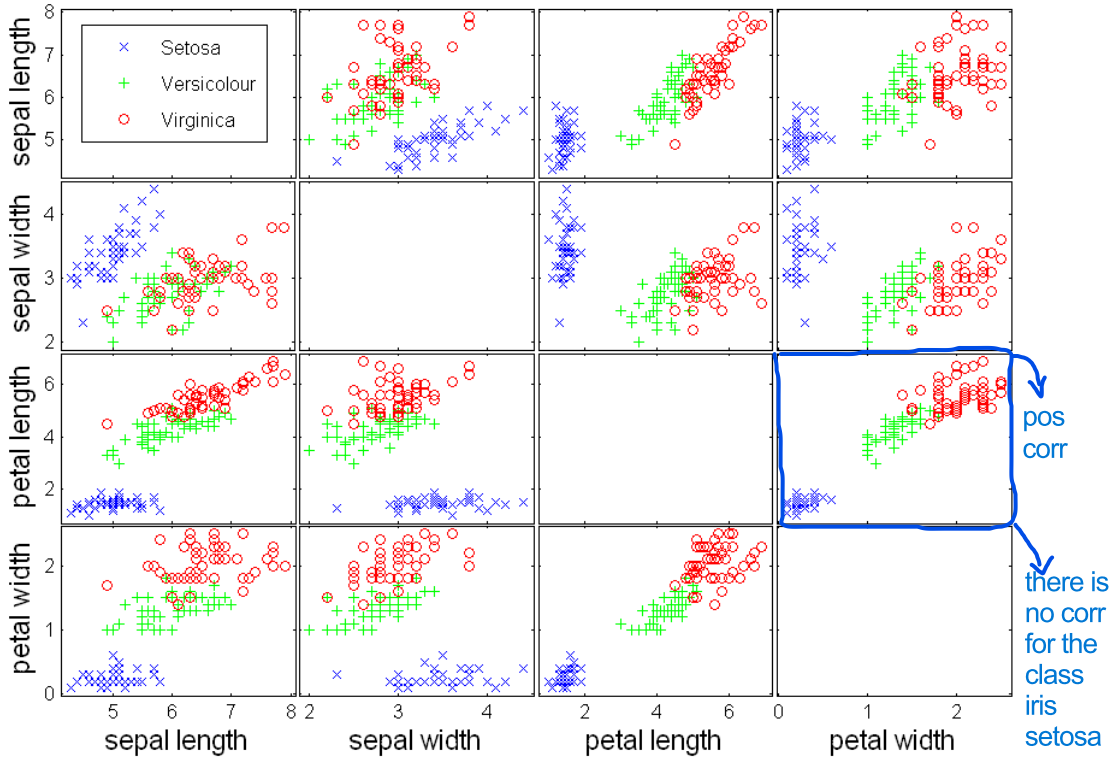
both increase

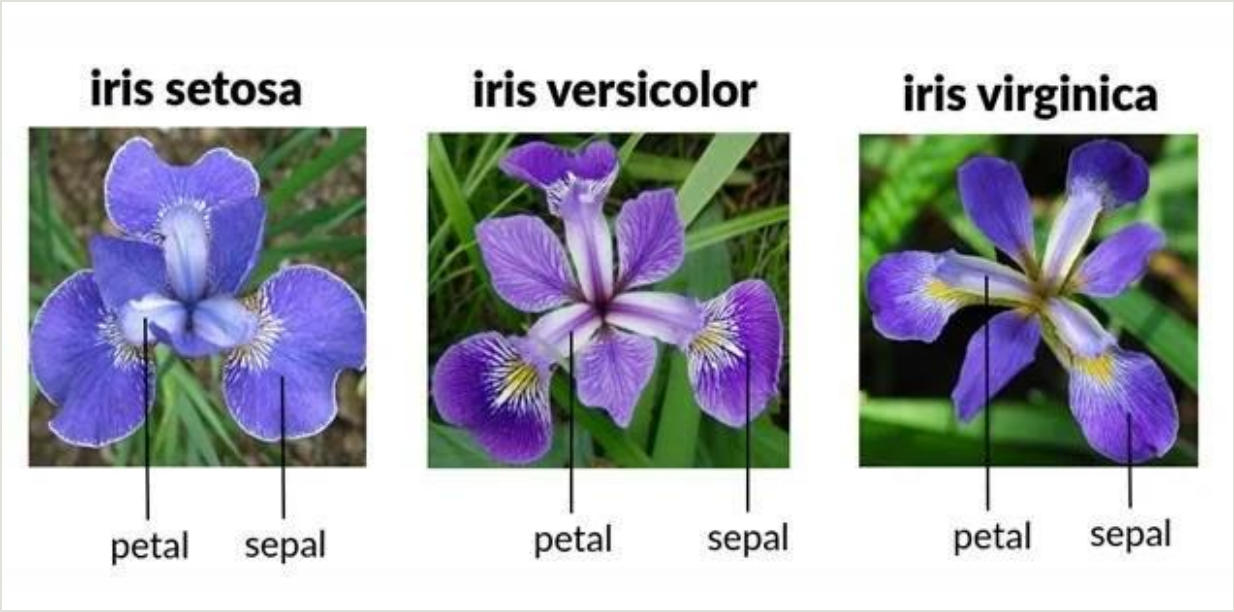
- ✓ Points are described by *x* and *y* continuous attributes.
- ✓ In each chart, attribute values were generated with different levels of correlation
- ✓ Is *x* values correlated to *y* ones?

lower he correla ion, higher is he blurring (no posi ioned in perfec line)

Visualizing Correlation: scatter plots

- Lets you determine whether some of the attributes are related
  - ✓ Useful for reducing the number of attributes to be considered in adistance/similatity function
- When labels are available, allows you to determine if you can classify objects based on the values of two attributes





Spearman Correlation

Pearson correlation is sensible to outliers. Spearman is robust to outliers (captures only strength of direction, not capable of capturing covariance of values)

- Spearman correlation measures the strength and direction of a **monotonic** relationship between two attributes: when x increases, y always increases (or always decreases), not necessarily linearly
  - ✓ It is the Pearson correlation computed on the ranks of the values instead of on the values themselves

$$\rho_s(\mathbf{x}, \mathbf{y}) = Corr\,(\,rank(\mathbf{x}),\ \ rank(\mathbf{y})\,)$$

- The rank of a value is its position in the list of the values sorted in ascending order (1 = smallest); tied values get the average of the positions they occupy
  - ✓ x = (10, 30, 20, 20, 50) → rank(x) = (1, 4, 2.5, 2.5, 5)

occupano seconda e terza posizione (quindi media dei due)

Saltare questa formula (focus sulla prima)

$$\rho_s(\mathbf{x}, \mathbf{y}) = 1 - \frac{6\sum_{k=1}^{n} d_k^2}{n\,(n^2 - 1)} \qquad d_k = rank(x_k) - rank(y_k)$$

- The second formula holds when there are no ties. ρ_s varies in [-1, 1]: 1 (-1) means that y is a monotonically increasing (decreasing) function of x

Spearman vs Pearson

When analyse the data, report the two correlations: if close together we have linear correlation, if not we need to investigate what is going on

- Spearman is preferable when:
  - ✓ The relationship is monotonic but not linear (exponential, logarithmic, saturation effects)
  - ✓ The data contain outliers: with ranks, the largest value always weighs n, whatever its magnitude
  - ✓ Distributions are strongly skewed (e.g., income, prices)
  - ✓ Attributes are ordinal (grades, satisfaction levels, rankings): only the order is meaningful, while Pearson requires meaningful differences, i.e., interval or ratio attributes
- Pearson is preferable when:
  - ✓ We are interested in linear relationships, e.g., to verify whether an attribute can be expressed as $x = a\,y + b$ and then discarded
  - ✓ Attributes are numerical and free of strong outliers: Pearson exploits the actual distances between values, not only their order
- Neither of them captures non-monotonic relationships: for $x = (-3, -2, -1, 0, 1, 2, 3)$ and $y = x^2$ both Pearson and Spearman are 0 → see Mutual Information

parabolic

Spearman Correlation: a Worked Example

- Study hours and exam grade of 6 students: the grade grows with the hours, with one exception (students S3 and S4)

Student	Hours x	Grade y	y=bx	rank(x)	rank(y)	d	d^2
S1	2	18	b=9	1	1	0	0
S2	4	20	b=5	2	2	0	0
S3	5	26	b=5.2	3	4	-1	1
S4	7	24	b=3,4	4	3	1	1
S5	8	28	b=3,5	5	5	0	0
S6	10	30	b=3	6	6	0	0

Correlation is just an information, but is not necessarily the truth

$\Sigma d^2 = 2, \quad n = 6$

$\rho_s = 1 - (6 \cdot 2) / (6 \cdot (36 - 1))$
$= 1 - 12 / 210 =$ **0.943**

Pearson on the same data = **0.923**

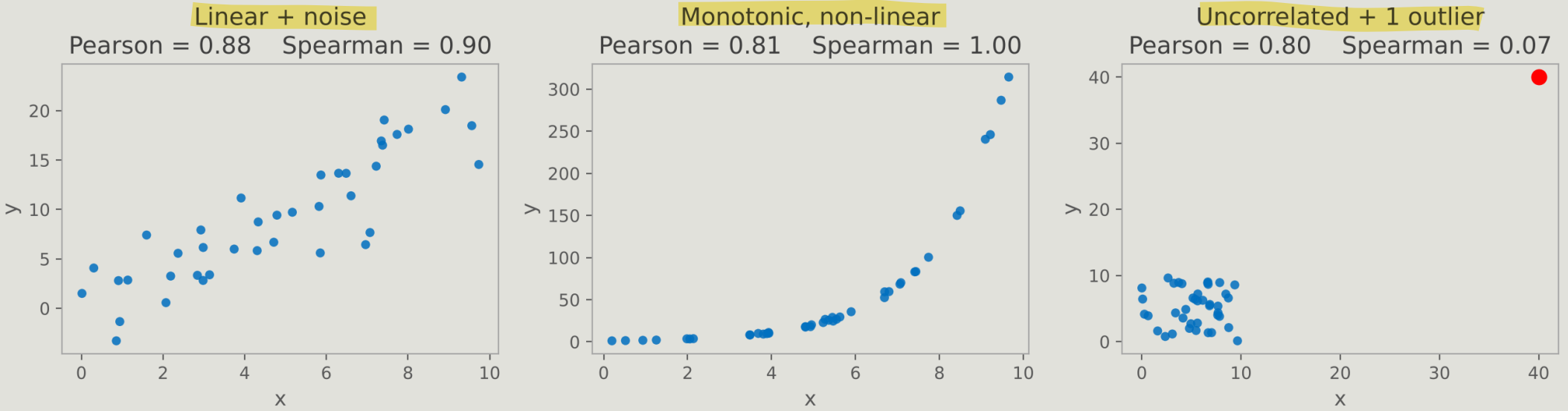
Both are high and close: the relationship is almost linear and there are no outliers

Aumentando linearmente le ore di studio, aumento non linearmente il voto (ho valori di b diversi per ogni studente: ciò mi suggerisce una non linearità della correlazione tra le due features)

monotonic and positive correlated but not linearly correlated

Cerca sempre di visualizzare i dati, ma basarsi solo sugli indicatori

Spearman vs Pearson: Visual Comparison



- ✓ Linear relationship with noise: the two coefficients are almost equal
- ✓ Monotonic but non-linear relationship: Spearman = 1 detects a perfect monotonic dependency, Pearson underestimates it because the data is not linear
- ✓ Uncorrelated data plus one extreme object (in red): a single outlier creates a strong Pearson correlation, Spearman is not fooled

a causa del fatto che usa la media dei valori

Spurious Correlation

- Spurious correlation: two attributes appear related (high correlation) but there is no direct cause–effect link between them
  - ✓ The association is really present in the data, but it is generated by something else: a hidden third factor, a common trend, chance, or the way data were collected
  - ✓ Correlation does not imply causation: a model can exploit a spurious correlation to predict, but it fails as soon as the hidden factor changes, and decisions based on it are wrong
- Typical sources
  - ✓ **Confounding variable**: a common cause z drives both x and y
  - ✓ **Common trend**: two time series that both grow (or decrease) over time, e.g., number of smartphones sold and average life expectancy
  - ✓ **Chance**: with many attributes some pairs are correlated just by chance (100 attributes → 4,950 pairs to test)
  - ✓ **Selection bias and aggregation**: data collected on a non-representative subset, or groups with different behavior merged together

Spurious Correlation: Formal Definition

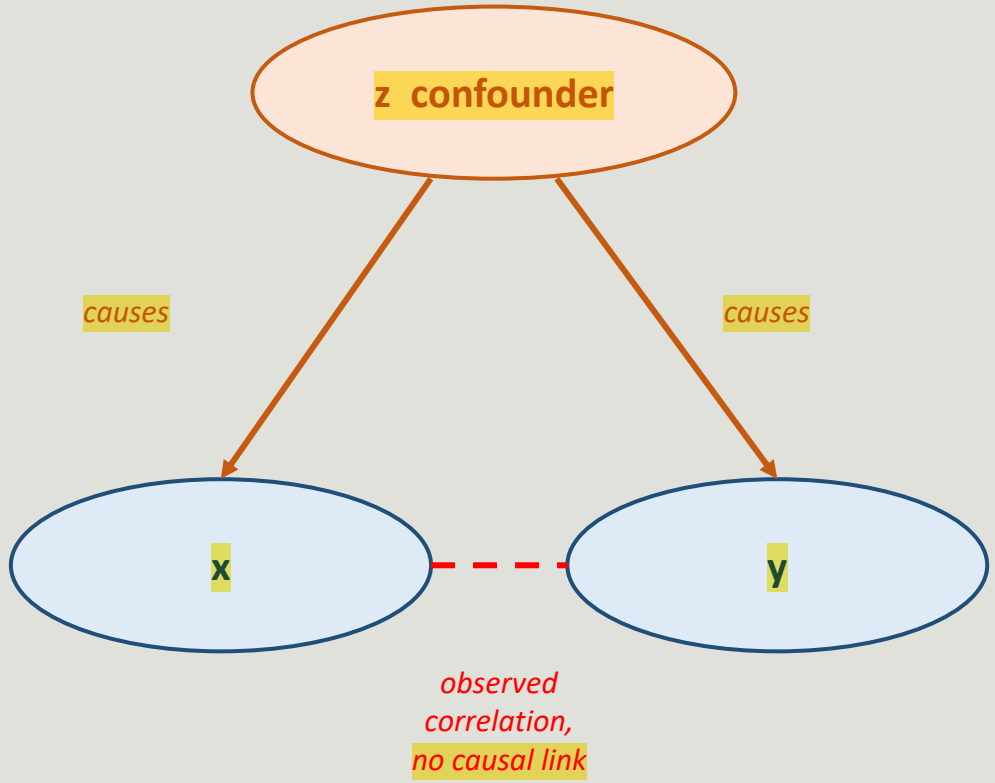
- **Formal definition**: x and y are spuriously correlated with respect to a variable z (confounder) when

$$\rho(\mathbf{x}, \mathbf{y}) \neq 0 \qquad \text{but} \qquad \rho(\mathbf{x}, \mathbf{y} \mid \mathbf{z}) \approx 0$$

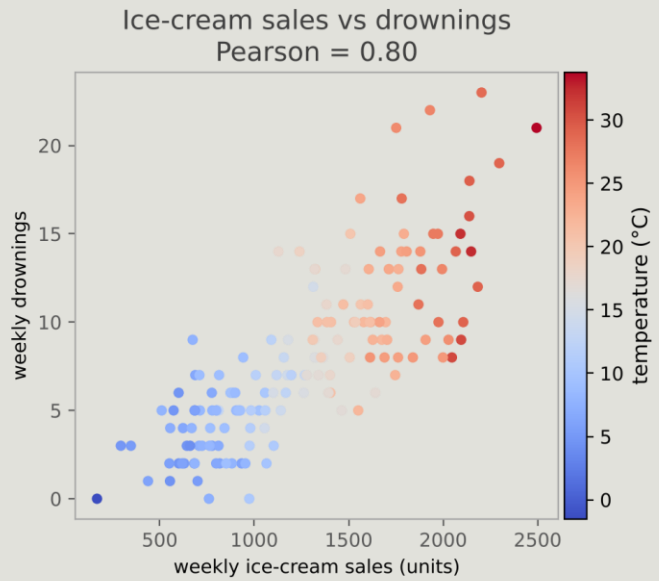
  - ✓ ρ(x, y | z) is the **partial correlation**: the correlation between x and y once the linear effect of z has been removed from both, i.e., the correlation between the residuals of x and y regressed on z

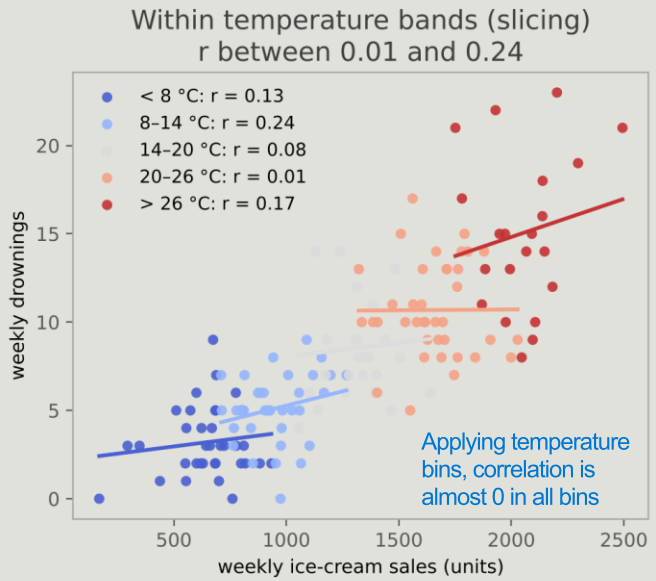
$$\rho(\mathbf{x}, \mathbf{y} \mid \mathbf{z}) = \frac{\rho_{xy} - \rho_{xz}\,\rho_{yz}}{\sqrt{(1-\rho_{xz}^2)\,(1-\rho_{yz}^2)}}$$

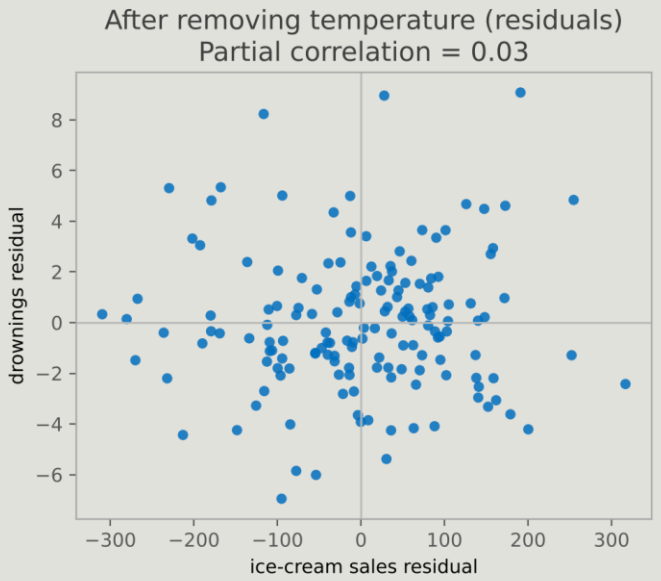
  - ✓ x and y are conditionally independent given z: once z is known, x carries no further information on y



Spurious Correlation: an Example







- ✓ Weekly ice-cream sales and drownings over 3 years (synthetic data) are strongly correlated: Pearson = 0.80
- ✓ Temperature is the ~~confounder~~ (hidden attribute): in hot weeks people buy more ice creams and more people go swimming, hence more drownings
- ✓ Within temperature bands, or after removing the effect of temperature, the correlation (almost) vanishes: banning ice cream would not save anyone

$\rho_{xy} = 0.80 \quad \rho_{xz} = 0.97 \quad \rho_{yz} = 0.82$

$\rho(x,y|z) = (0.80 - 0.97 \cdot 0.82) / \sqrt{(1 - 0.97^2)(1 - 0.82^2)} \approx \mathbf{0.03}$

Detecting Spurious Correlation

- Domain knowledge: is there a plausible mechanism linking x and y? Verify the hypothesis with the domain experts (see Business Understanding)
- Look for confounders: list candidate common causes (time, season, location, size of the object, …) and control for them
  - ✓ Dataset slicing / stratification: compute the correlation within groups where z is (almost) constant; if it vanishes, the correlation is spurious
  - ✓ Partial correlation: regress x and y on z and correlate the residuals
- Time series: remove trends and seasonality (e.g., by differencing) before computing the correlation
- Chance: with many attributes, check the statistical significance of the correlation (correcting for multiple comparisons) and confirm it on new data

Correlation is a hint, not a proof: these checks can reveal that a correlation is spurious, but only controlled experiments (randomization, A/B tests) can establish a causal relationship

Indicator of correlation (provide information about all type of correlation: not only linear one)

Mutual Information

- **Mutual information** measures how much knowing one variable helps us predict the other.
- For two discrete random variables ***x*** and ***y***, with joint probability function $p(\boldsymbol{x},\boldsymbol{y})$ and marginal probability functions $\boldsymbol{p}(\boldsymbol{x})$ and $\boldsymbol{p}(\boldsymbol{y})$, the mutual information is defined as:

prob that appear together

prob that appear without considering the other variable

$$\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) = \sum_{x\in\boldsymbol{x}}\sum_{y\in\boldsymbol{y}} p(x,y)\log\frac{p(x,y)}{p(x)\cdot p(y)}$$

  - $\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) \geq 0$
  - $\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) = 0$ iff **x** and **y** are independent (i,.e. knowing the value of **x** does not provide any information about the **y** value)
  - $\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) = \mathrm{MI}(\boldsymbol{y},\boldsymbol{x})$ symmetry
- Mutual information captures all the types of correlation (not only the linear one)
- Definition for continuous variables is technically different but the concepts are the same

Mutual Information

Quando si parla di popolazione, p indica probabilità
Quando si parla di dataset, p indica frequenza

Suppose we have two binary variables:

- ***x*** = whether it's raining (Yes/No)
- ***y*** = whether people carry umbrellas (Yes/No)

Let's say:

- p(***x*** = Yes) = 0.3 x=0.3 esce il 30% delle volte nel dataset (3 volte su 10), qualsiasi sia il valore di y
- p(***y*** = Yes) = 0.2 y=0.2 esce il 20% delle volte nel dataset (2 volte su 10), qualsiasi sia il valore di x

If independent, p(***x*** = Yes, ***y*** = Yes) = 0.3∗0.2 = 0.06
joint probability

Now in reality:

- p(***x*** = Yes, ***y*** = Yes) = 0.15 > 0.06
joint probability

Interpretation: **seeing rain makes carrying an umbrella more likely** → dependency → positive MI contribution.

Mutual Information

- **Mutual information** measures how much knowing one variable helps us predict the other.
- For two discrete random variables ***x*** and ***y***, with joint probability function $p(\boldsymbol{x},\boldsymbol{y})$ and marginal probability functions $\boldsymbol{p}(\boldsymbol{x})$ and $\boldsymbol{p}(\boldsymbol{y})$, the mutual information is defined as:

$$\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) = \sum_{x\in\boldsymbol{x}}\sum_{y\in\boldsymbol{y}} p(x,y)\log\frac{p(x,y)}{p(x)\cdot p(y)}$$

- $\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) \geq 0$

Joint probability $p(x,y)$
- Acts as a weight in the sum. (if it is more frequent, higher; if it is rare, lower weight)
- Events that are more likely (higher $p(x,y)$) contribute more to MI.
- Rare events (small p(x,y)p(x, y)p(x,y)) contribute less.
- Intuition: "Focus on what actually happens often."

ng the value of **x** does not provide

not only the linear one)

but the concepts are the same

Mutual Information

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- $\mathrm{MI}(\boldsymbol{x},\boldsymbol{y}) \geq 0$

Product of marginals $p(x)\cdot p(y)$

- Represents what would happen if ***x*** and ***y*** were independent.
- If ***x*** and ***y*** are independent, $p(x,y) = p(x)\cdot p(y)$, and the log term becomes 0.
- Intuition: this is your **baseline expectation** for each combination — deviation from this is information.

log(1) = 0

Comparing $p(x) \cdot p(y)$ to $p(x, y)$

- If $p(x, y) > p(\mathrm{x})\, p(\mathrm{y})$ (so not independent: there is some relation)
  - The pair (*x*,*y*) occurs more often than we would expect if ***x*** and ***y*** were independent.
  - This means that seeing ***x***=*x* increases the likelihood of ***y***=*y* (or vice versa).
- Conversely, $p(x, y) < p(\mathrm{x})\, p(\mathrm{y})$
  - The pair occurs less often than expected under independence.
  - Seeing seeing ***x***=*x* makes ***y***=*y* less likely than random chance.
- Both cases indicate **statistical dependence**, because the probability of one variable depends on the value of the other.

probability functions ***p***(***x***) and ***p***(***y***), the mutual informat

$$\mathrm{MI}(\boldsymbol{x}, \boldsymbol{y}) = \sum_{x \in \boldsymbol{x}} \sum_{y \in \boldsymbol{y}} p(x, y) \log \frac{p(x, y)}{p(x) \cdot p(y)}$$

- $\mathrm{MI}(\boldsymbol{x}, \boldsymbol{y}) \geq 0$
- $\mathrm{MI}(\boldsymbol{x}, \boldsymbol{y}) = 0$ iff **x** and **y** are independent (i,.e. knowing the value of **x** does not provide any information about the **y** value)
- $\mathrm{MI}(\boldsymbol{x}, \boldsymbol{y}) = \mathrm{MI}(\boldsymbol{y}, \boldsymbol{x})$ symmetry

- Mutual information captures all the types of correlation (not only the linear one)
- Definition for continuous variables is technically different but the concepts are the same

Mutual Information

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- For two discrete random variables ***x*** and ***y***, with joint probability function $p(\boldsymbol{x},\boldsymbol{y})$ and marginal probability functions $\boldsymbol{p}(\boldsymbol{x})$ and $\boldsymbol{p}(\boldsymbol{y})$, the mutual information is defined as:

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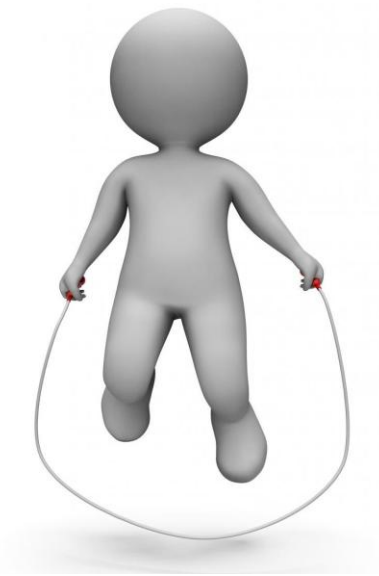
The **log** in mutual information does more than just "make MI zero when independent i.e. (p(x,y) = p(x) p(y) = 1

The log transforms multiplicative "*ratio*" into **additive information**.

- A ratio of **2** means it is twice as likely.
- A ratio of **0.5** means it is half as likely.

The ratio tells us a **factor of change**, not an information amount. Averaging ratios 2 and 0.5 do not return 1, whiel their base 2 logarithms: +1 and −1, express them as equal-sized contributions in opposite directions.

The Titanic Dataset



[The kaggle dataset]

The sinking of the Titanic is one of the most infamous shipwrecks in history.

On April 15, 1912, during her maiden voyage, the widely considered “unsinkable” RMS Titanic sank after colliding with an iceberg. Unfortunately, there weren’t enough lifeboats for everyone on board, resulting in the death of 1502 out of 2224 passengers and crew.

While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.



Multivariate Analysis

Humans cannot interpret high-dimensional spaces

- Hard to visualize beyond 3 dimensions
- Difficult to understand relationships

Dimensionality reduction enables 2D/3D visualization

They can be used even in Feature selection and in modeling (Now, our goal is Data understanding)

- Principal Component Analysis (PCA)
  - Linear transformation of the multi dimensional space
  - Maximizes global variance
  - Preserves Keeps overall data structure while projecting multi-dimensional data to a 2D/3D space
  - Distances are meaningful at a global scale
- t-distributed Stochastic Neighbor Embedding (T-SNE)
  - Nonlinear transformation
  - Preserves local neighborhoods (ciò che è vicino viene preservato, ma non viene preservato ciò che è a distanza lontana)
  - Points close in high dimensions stay close in low dimensions
  - Global distances are not reliable

Dimensionality Reduction: beyond Data Understanding

As the number of variables grows, data becomes sparse and distances loose meaning—this is known as the Curse of Dimensionality. (so, we need to reduce dimensionality when possible)

- Points tend to look equally distant from each other
- Neighborhood-based methods (e.g., clustering) degrade
- Models require exponentially more data

In real datasets, many variables are correlated or redundant. Techniques like PCA transform correlated variables into uncorrelated components, preserving most of the information.

- Example: income, spending score, and credit limit often overlap
- Keeping all variables adds noise without adding information

High-dimensional data can cause:

- Overfitting (model learns noise)
- Instability in parameter estimation

Reducing dimensions restores structure and improves learning.

We will exploit them again during the feature engineering step.

Principal Component Analysis

combination of previous axes of high dimentionality to get maximum variance while orthogonal

PCA is a **linear dimensionality reduction technique** grounded in **linear algebra and statistics**. Its goal is to transform a dataset into a new coordinate system where the axes (principal components) capture the **maximum variance** in decreasing order, while being **orthogonal** (uncorrelated).

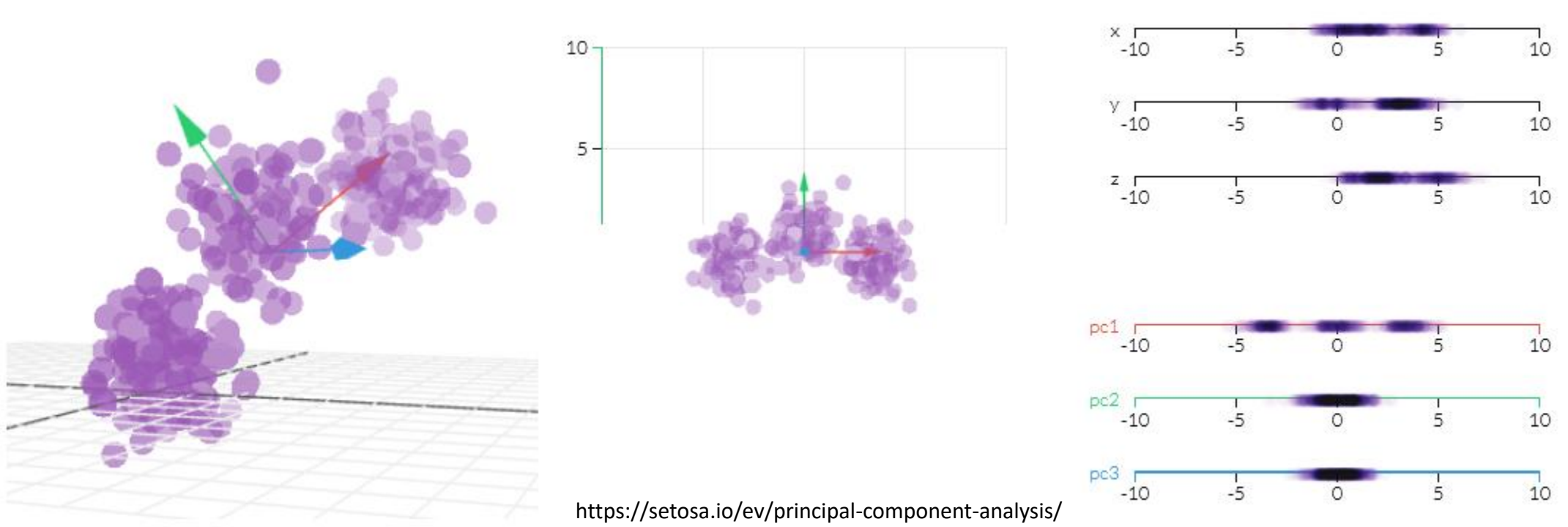
PCA is based on the following assumptions (not necessarily the truth: when both assumptions are false, PCA is less effective)

- **Information–variance assumption:** high-variance directions contain the most relevant information.
  - High variance → data points are well spread → rich structure
  - Low variance → data points are compressed → little information
- **Linearity assumption:** the relevant data structure can be captured through a global linear transformation and represented by a lower-dimensional linear subspace.

So when PCA maximizes variance, it is effectively: Finding directions that preserve as much information as possible

Principal Component Analysis

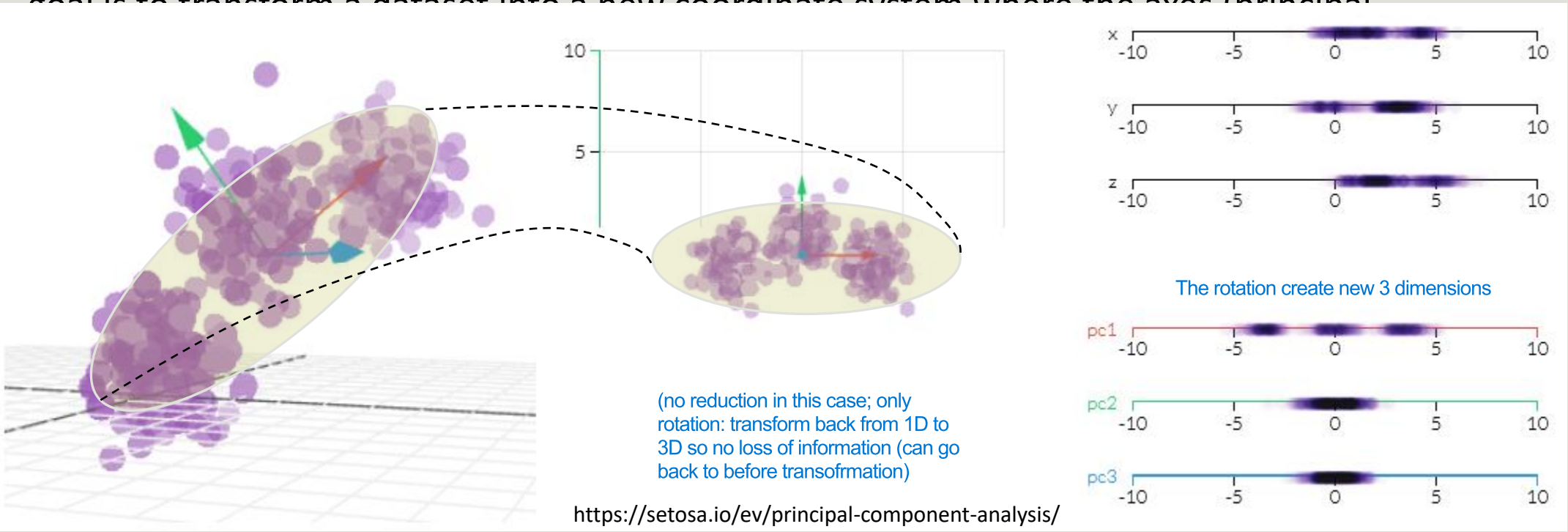
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https://setosa.io/ev/principal-component-analysis/

Principal Component Analysis

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reduction if i lose the pc3 and maintain pc1 and pc2

Linear Projectection

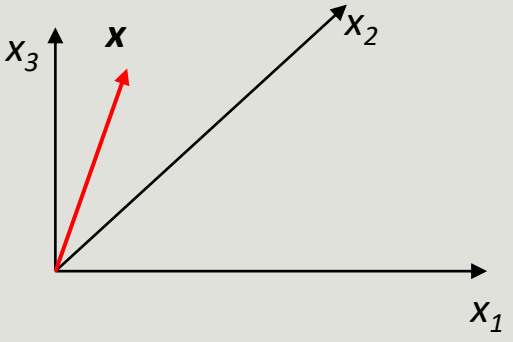
Let $\boldsymbol{x}$ a 3D point be:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Given two linearly independent vectors (i.e. different directions) in 3D defining the projection plane:

$$\boldsymbol{p}_1 = (p_{11}, p_{12}, p_{13}) \qquad \boldsymbol{p}_2 = (p_{21}, p_{22}, p_{23})$$

Then the projected 2D point is: $\boldsymbol{y} = \boldsymbol{P}^T\boldsymbol{x}$ where $\boldsymbol{P} = \begin{bmatrix} p_{1,1} & p_{2,1} \\ p_{1,2} & p_{2,2} \\ p_{1,3} & p_{2,3} \end{bmatrix}$



Carrying out the matrix multiplication:

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} p_{1,1}x_1 + p_{1,2}x_2 + p_{1,3}x_3 \\ p_{2,1}x_1 + p_{2,2}x_2 + p_{2,3}x_3 \end{bmatrix}$$

Each 2D coordinate is a **linear combination** of the original 3D coordinates.

Linear Projectection

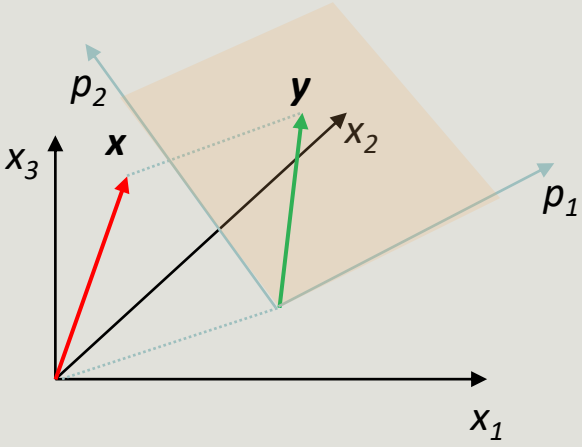
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Each 2D coordinate is a **linear combination** of the original 3D coordinates.

Mathematical Formulation for PCA

$$Cov(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{n}\sum_{k=1}^{n}(x_k - \bar{x})(y_k - \bar{y})$$

Which is the projection from the original dataset space R^n to R^k with $k<n$ that summarize data at best?

Data distribution is properly described by the covariance matrix $\boldsymbol{\Sigma}=\frac{1}{n}[(\boldsymbol{x}-\boldsymbol{\mu})(\boldsymbol{x}-\boldsymbol{\mu})^T]$ in R^n

We need to find the R^k space where the projected variance (i.e. data distribution) is maximized

This can be solved as a constrained optimization problem that maximizes the variance of the projected data called **Maximum variance projection** or **Rayleigh quotient maximization**

coviariance matrix: how all the variable vary each other

The solution is given by the transformation $\boldsymbol{V}$ determined by the **eigenvectors of the covariance matrix** $V = [\mathbf{V}_1,\ldots \mathbf{V}_n]$

- Eigenvectors $\boldsymbol{V}$ are directions that remain invariant (up to sign) under a linear transformation
- In PCA, eigenvectors of the covariance matrix define a new **orthogonal coordinate system**
- The factor by which they are scaled (due to the data distribution) is called the eigenvalue $\lambda=[\lambda_1, \lambda_2,\ldots, \lambda_n]$

how much the information is spread along the new dimensions

Each eigenvalue λ_i tells you how much variance lies along its eigenvector (i.e. how stretched the cloud is along each axis)

- Large eigenvalue → important direction
- Small eigenvalue → less important

Mathematical Formulation for PCA

The data distribution is exactly preserved under eigenvector transformation (just expressed in a new coordinate system).

- The new coordinate system determined by eigenvectors does not apply any dimensionality reduction!
- Data are still in R^n but axes changed. We carried out a change of basis in R^n

To carry out dimensionality reduction we

1. Order the eigenvectors $\mathbf{V}_i$ in descending order of the corresponding eigenvalues $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_n)$ with $\lambda_1 > \lambda_2 > \cdots > \lambda_n$
2. We pick the first *k* eigenvectors that explain enough variance
3. Given the percentage *t* of the informative content to be preserved (e.g. *t* = 95%) we select the minimum number of eigenvector *k* such that the sum of the largest eigenvalues, with respect to the sum of all the eigenvalues, is higher than or equel to *t*.

$$k = \text{argmin}_z \left(\frac{\sum_{i=1..z} \lambda_i}{\sum_{i=1..n} \lambda_i} > t \right)$$

Projecting and Retro-Projecting the Data

Projecting the data on the first *k* eigenvector $\boldsymbol{z_k} = \boldsymbol{V}_k^T \boldsymbol{x}$

- Data are represented in R^k data space with $k<n$
- Only the main components of the datasets are preserved

Retro-projecting from the reduced to the original data space $\tilde{\boldsymbol{x}} = \boldsymbol{V}_k \, \boldsymbol{z_k}$

- Data are represented in the original R^n data space
- The original and the retropropagated points are different ($\tilde{\boldsymbol{x}} \neq \boldsymbol{x}$) since "details" are lost in projection
  - Two distinct points $\boldsymbol{x},\boldsymbol{y} \in R^n$, that have the same first *k* eigenvalues, will be projected to the same point in $\boldsymbol{z} \in R^k$ and retroprojected to the same point in $\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{y}} \in R^n$

Reading the Components: the Weights

Weights: each principal component is a linear combination of the original dimensions

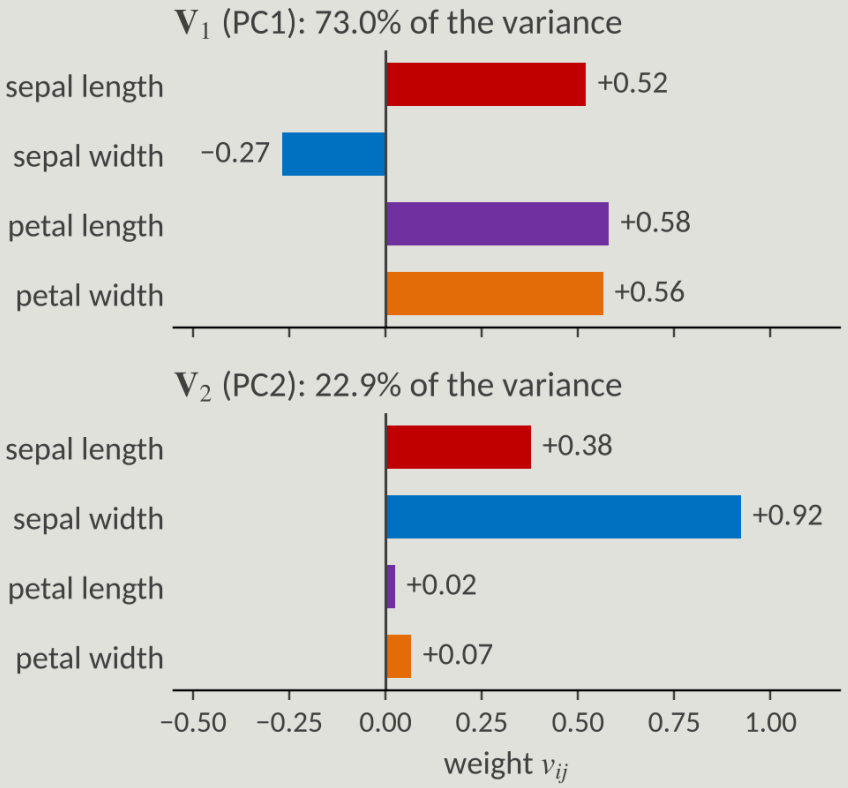
$$z_j = V_j^T \boldsymbol{x} = v_{1j}x_1 + v_{2j}x_2 + \ldots + v_{nj}x_n$$

- v_{ij}, the i-th element of the eigenvector V_j, is the **weight** of the original dimension x_i in the component z_j
- **Sign**: + → x_i and z_j grow together; − → they move in opposite directions
- **Magnitude**: since $\|V_j\| = 1$, then $\Sigma_i\, v_{ij}^2 = 1$, so v_{ij}^2 is the share of z_j due to x_i
- **Loading**: $v_{ij} \cdot \sqrt{\lambda_j}$ is the correlation between x_i and z_j (standardized data)

Standardize the data first: otherwise the weights reflect the units of measure and the dimension with the largest variance dominates the first component

Example: Iris dataset, 4 standardized dimensions

- ✓ PC1 ≈ **flower size**: petal length, petal width and sepal length have similar positive weights
- ✓ PC2 ≈ **sepal width**: the only dimension with a large weight (0.92)



Reading the Components: the Biplot

Biplot: a scatter plot of the reduced space R^k that shows together

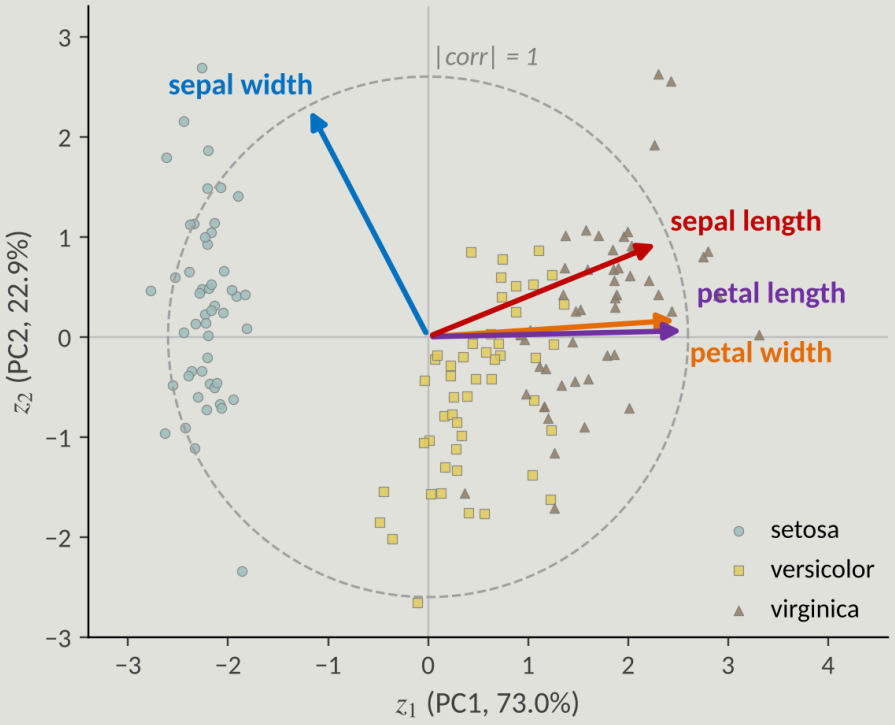
- **the points**, i.e., the projected data $z = V_k^T x$
- **the original dimensions**, one arrow for each x_i

Where do the arrows come from? The unit vector e_i of the original axis x_i is projected as $V_k^T e_i = (v_{i1}, \ldots, v_{ik})$: the i-th row of V_k, i.e., the weights of x_i in the k components (usually drawn as loadings, i.e., correlations)

How to read it

- **Direction**: where x_i grows; points along the arrow have high values of x_i, points on the opposite side low values
- **Length**: how well x_i is represented in R^k (length² = share of its variance explained); a short arrow → x_i lies mostly in the discarded components
- **Angle** between two arrows: its cosine ≈ correlation (0° positive, 90° uncorrelated, 180° negative)

✓ Iris: petal length and width overlap (corr. 0.96) and point to *virginica*; sepal width forms an angle of ≈116° with them: cos = −0.43 = their correlation



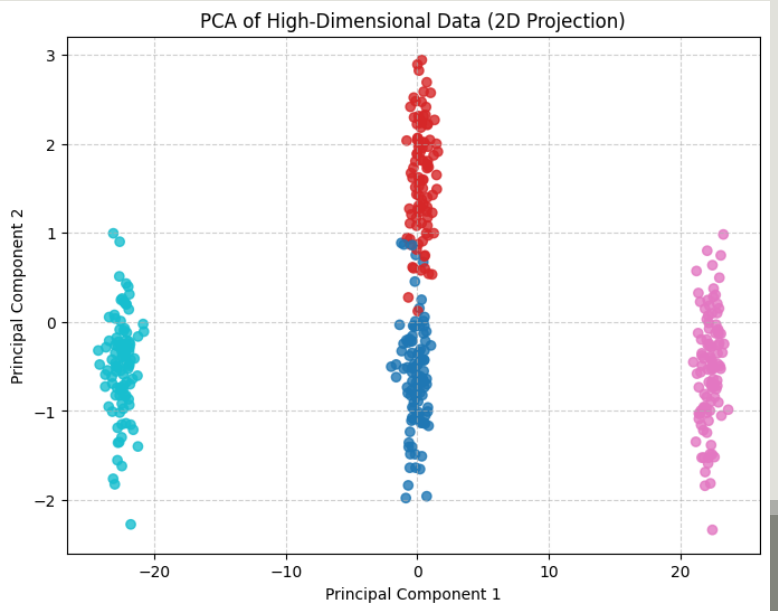
PCA at Works

1) Set 4 centroids in a 5-dimension dataspace
2) Generate a 400 points dataset

```
[0, 0, 0, 0, 0],    # blob 1
[1, 1, 1, 1, 1],    # blob 2 (close to blob 1)
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[-10, -10, -10, 10, 10]    # blob 4 (far away)
```

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[-10.79201351 -10.36706145 -10.02222208   9.74241867   9.58454741]
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[ -0.53370866  -0.48948617  -0.04626103   0.20469118   0.16601448]
 ...
[ -9.97383853  -8.98296938 -10.37358962  10.11676447   9.55451765]
[  0.0147061    0.29879897   0.87068616   0.5755625    1.29190947]
[-10.8745309  -10.18552545 -10.45129384  10.19150471  10.80427027]
```

3) Apply PCA to 2D

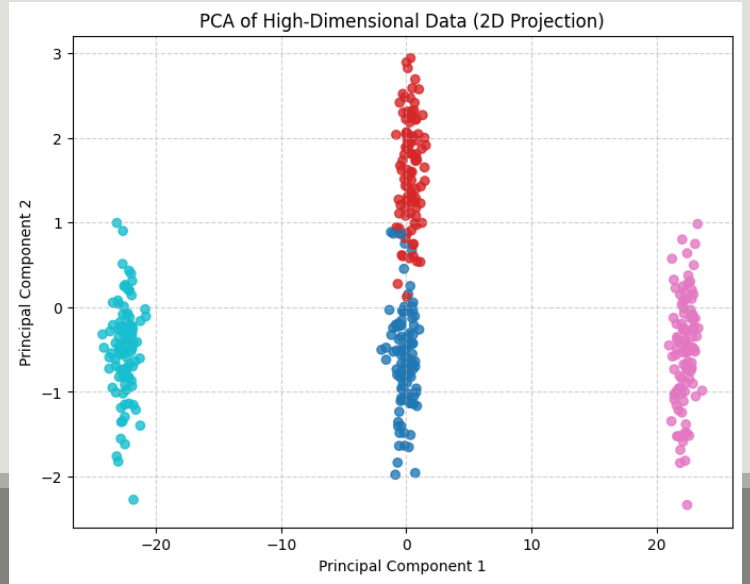


When Does PCA Fail?

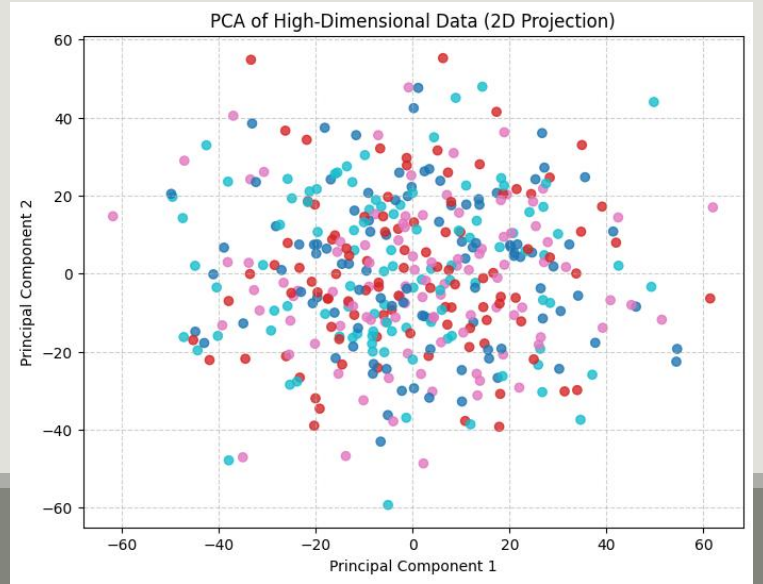
PCA approximates the data through a linear subspace identified from the covariance structure. It is particularly effective when the data form an approximately elongated, ellipsoidal cloud around that subspace. In many cases this is not true

- Principal components summarize variance and linear dependence, but may fail to preserve nonlinear or local characteristics of the data distribution.
- Banana-, circular-, spiral-shaped datasets

In all these cases PCA loses relevant information and may yield misleading results. In such cases, a nonlinear transformation is necessary.



Adding 2 high variance random features to the original dataset

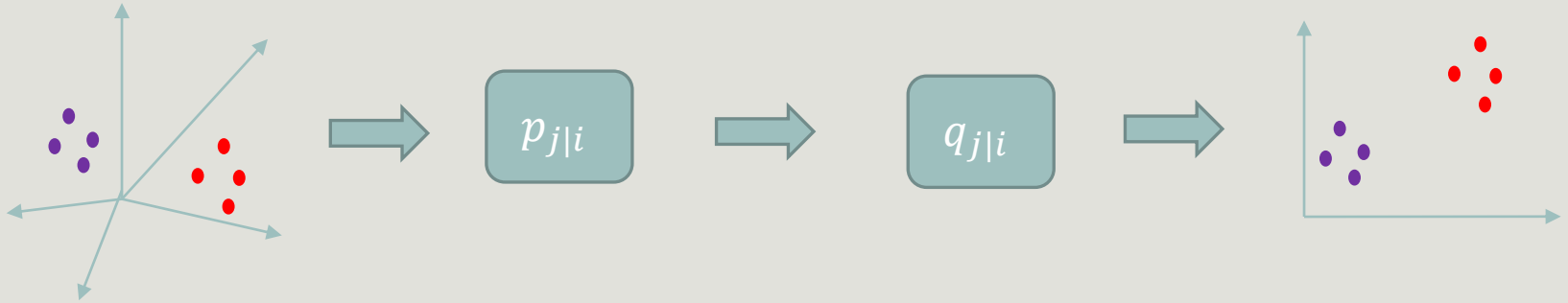


t-Distributed Stochastic Neighbor Embedding

t-SNE is a **nonlinear dimensionality reduction** method that preserves **local neighborhood structure**.

- Mainly used to visualize high-dimensional data in 2D, 3D
- Reveals clusters and local neighborhoods: local structure is preserved, not global geometry
- Handles non-linear structure
- t-SNE is not linear and does not preserve global geometry because it does not apply a single global transformation (like a matrix multiplication); instead, it *iteratively rearranges points* ~~to match pairwise similarities~~ in order to preserve local (pairwise) similarities using a nonlinear objective.

Defining probability distributions over pairs of points in both high- and low-dimensional spaces, and minimizing their mismatch by letting the system learn ***Q*** from ***P***.



How t-SNE work?

Non andare oltre nel dettaglio di spiegazione

1) Converts distances in high-dimensional spaces in probabilities *P*: Si vuole rappresentare le distanze come probabilità

- For each of the *N* point $\boldsymbol{x}_i$, it defines a probability distribution over all other points using a *Gaussian kernel*

Probabilità della coppia dei punti j-i

$$p_{j|i} = \frac{\exp(-\left||x_i - x_j|\right|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\left||x_i - x_k|\right|^2 / 2\sigma_i^2)}$$

- Then symmetrizes ***P***: $p_{j|i} = \frac{p_{j|i} + p_{i|j}}{2N}$
- Nearby points get high probability; distant points get near-zero.

2) Define a similar distribution *Q* in low dimensions using a *t-student kernel* $q_{ij} = \frac{(1+\left||y_i - y_j|\right|^2)^{-1}}{\sum_{k \neq l}(1+\left||y_k - y_l|\right|^2)^{-1}}$

Iteratively,

3) Let the system learn the *Q* probability distribution by ~~minimizing the Kullback–Leibler divergence~~ optimization problem

$$KL(P||Q) = \sum_{i \neq j} p_{ij} log \frac{p_{ij}}{q_{ij}}$$

- KL is minimized when $p_{ij} \approx q_{ij}$

(Minimizza la differenza di probabilità tra quella nello spazio ad alta dimensionalità e quella nello spazio a bassa dimensionalità: due punti hanno alta probabilità in entrambi gli spazi e quindi sono vicini)

σ_i^2 varies for each point x_i depending on the **perplexity parameter** that smoothly defines **the "effective number of neighbors"** each point considers.

- **Low perplexity (e.g. 5–10):** Emphasizes very local structure and can produce **fragmented clusters**
- **Medium perplexity (e.g. 20–30):** Balanced view of local clusters
- **High perplexity (e.g. 40–100):** Captures broader/global structure but can merge small clusters

A **kernel** is a rule that turns the distance between two points into a similarity value: close points get high similarity, distant points get low similarity.

probabilities *P*:

n over all other points using a *Gaussian kernel*

$$p_{j|i} = \frac{\exp(-\left||x_i - x_j|\right|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\left||x_i - x_k|\right|^2 / 2\sigma_i^2)}$$

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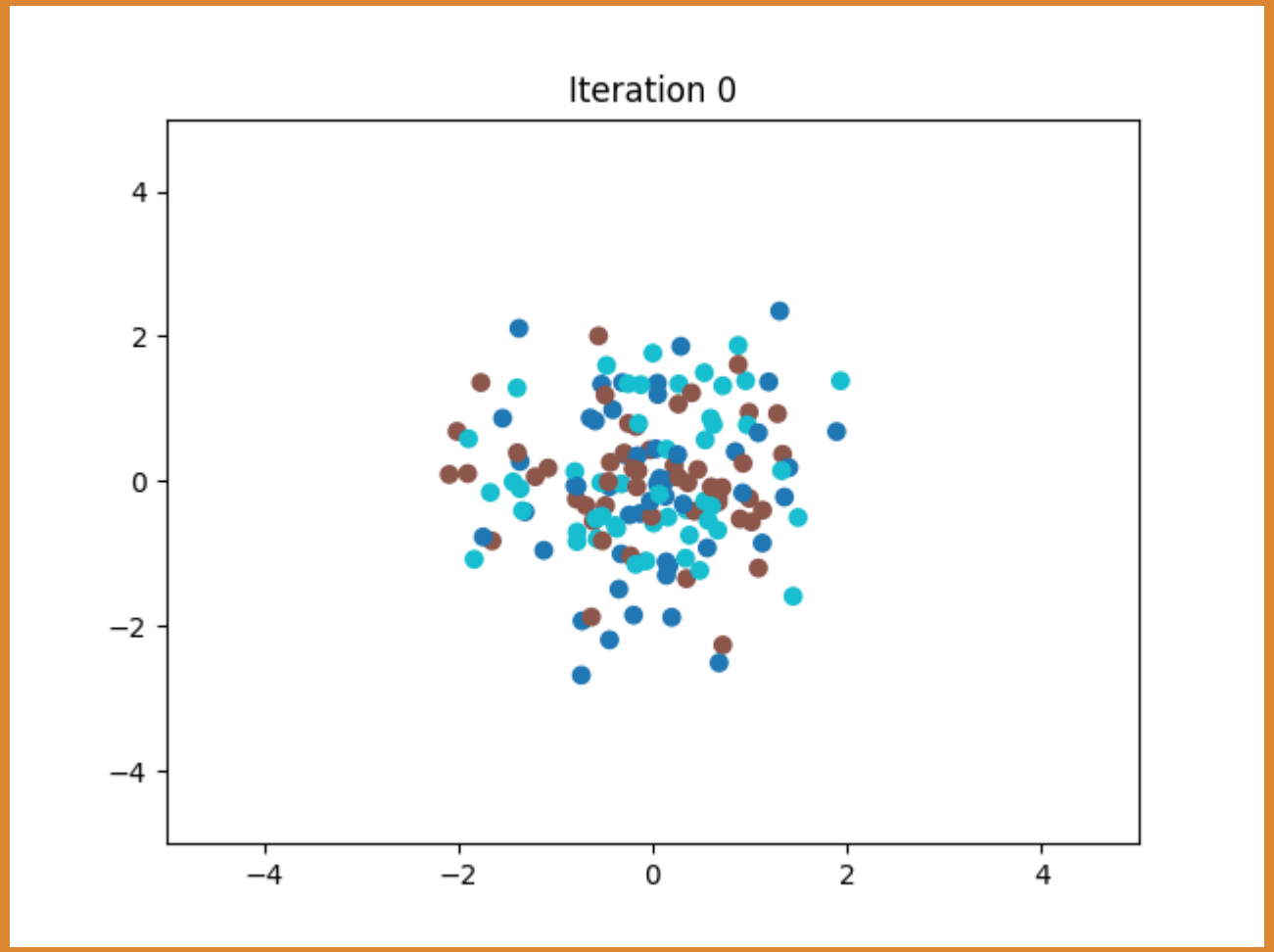
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3) Let the system learn the *Q* probability distribution by minimizing the Kullback–Leibler divergence

$$KL(P||Q) = \sum_{i \neq j} p_{ij} log \frac{p_{ij}}{q_{ij}}$$

- KL is minimized when $p_{ij} \approx q_{ij}$

Learning ***Q*** for a 5 dimensional dataset with three clusters



kernel

$$\frac{\left\|y_j\right\|^2)^{-1}}{\left\|y_l\right\|^2)^{-1}}$$

vergence

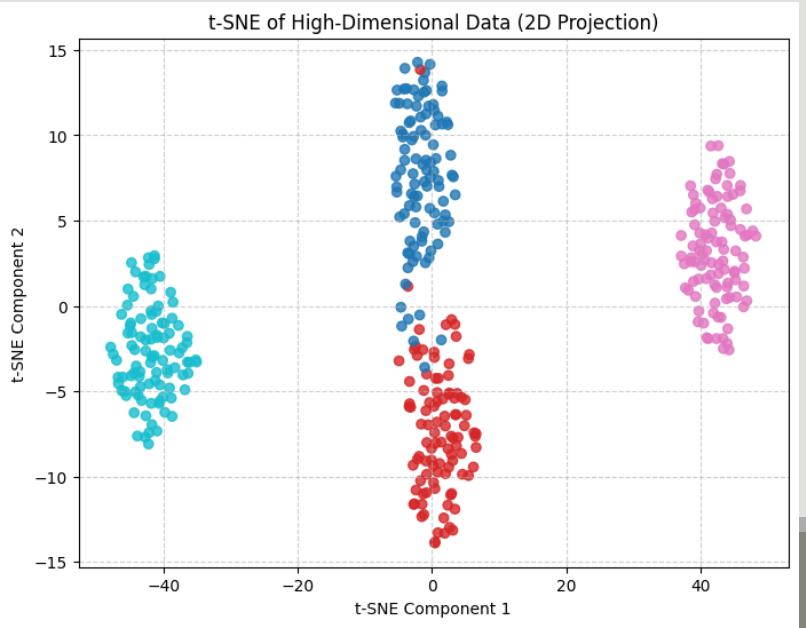
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[ -0.53370866  -0.48948617  -0.04626103   0.20469118   0.16601448]
 ...
[ -9.97383853  -8.98296938 -10.37358962  10.11676447   9.55451765]
[  0.0147061    0.29879897   0.87068616   0.5755625    1.29190947]
[-10.8745309  -10.18552545 -10.45129384  10.19150471  10.80427027]
```

3) Apply T-SNE to have a 2D representation

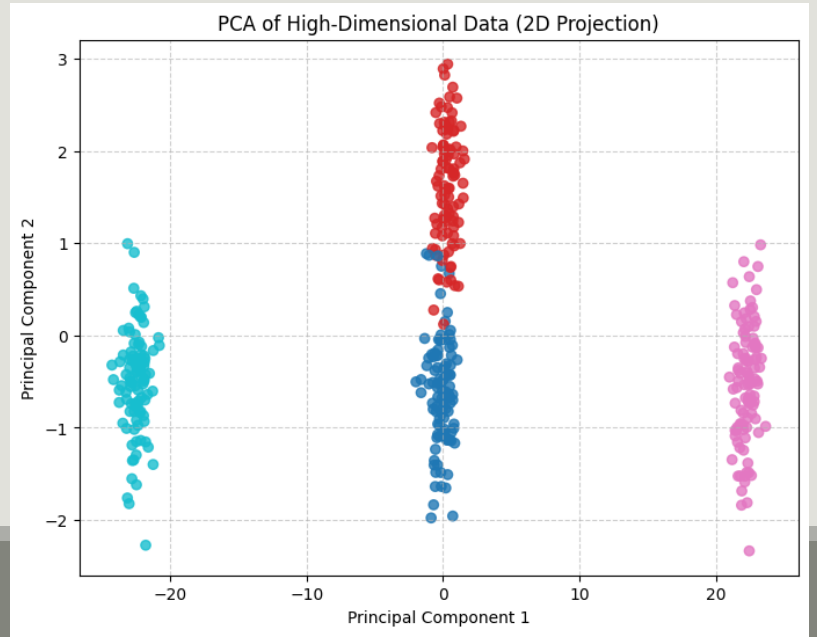


T-SNE at Works

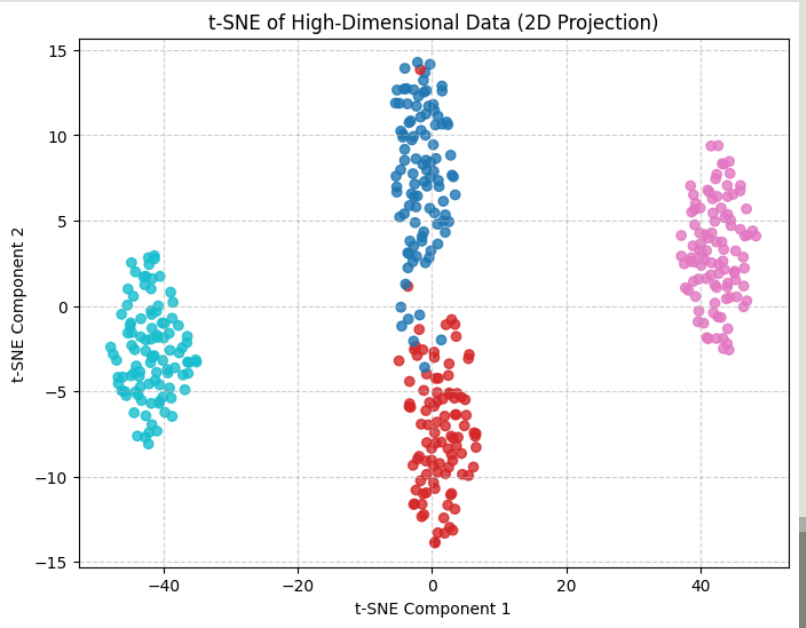
1) Set 4 centroids in a 5-dimension dataspace
2) Generate a 400 points dataset

```
[0, 0, 0, 0, 0],     # blob 1
[1, 1, 1, 1, 1],     # blob 2 (close to blob 1)
[10, 10, 10, -10, -10],     # blob 3 (far away)
[-10, -10, -10, 10, 10]     # blob 4 (far away)
```

```
[-10.79201351 -10.36706145 -10.02222208   9.74241867   9.58454741]
[  0.31160791   0.91964335  -0.06525609   0.24102703   0.4140864 ]
[ -0.53370866  -0.48948617  -0.04626103   0.20469118   0.16601448]
 ...
[ -9.97383853  -8.98296938 -10.37358962  10.11676447   9.55451765]
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```



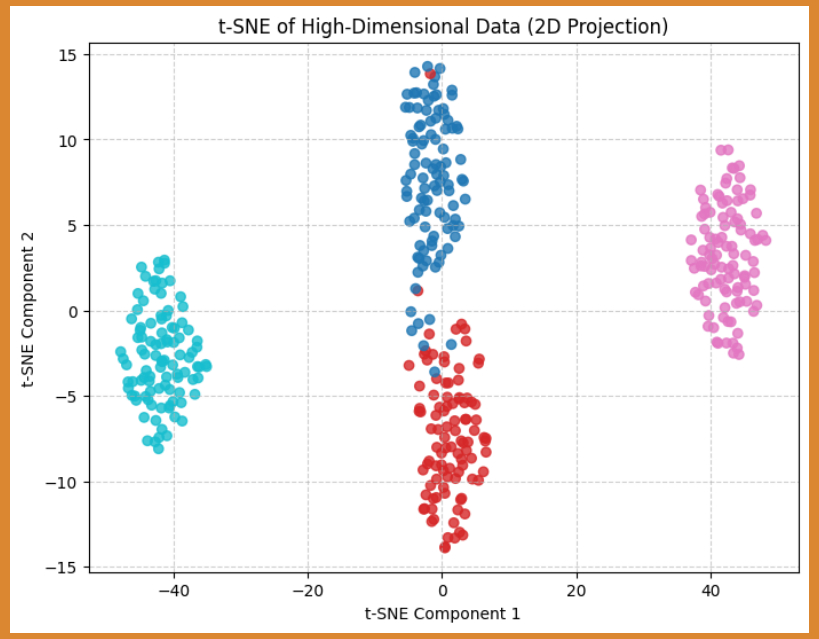
PCA ≈ T-SNE



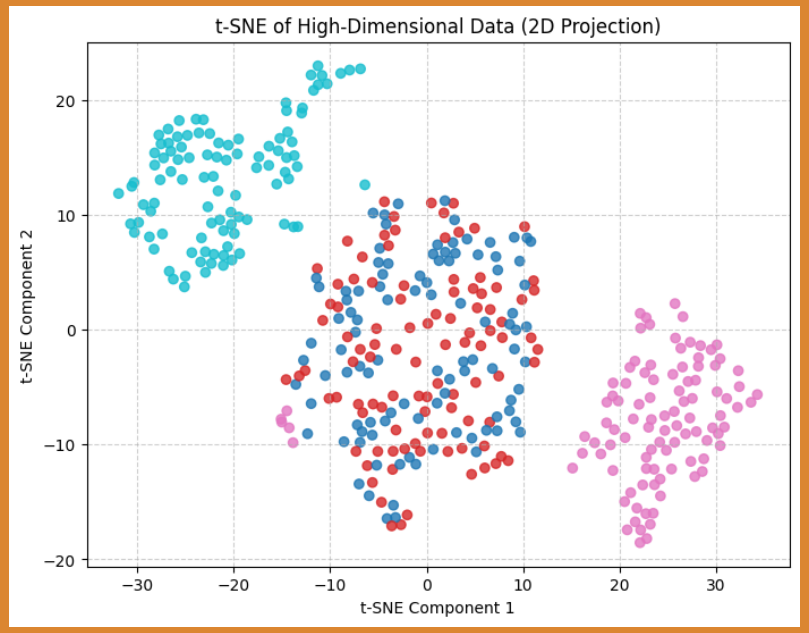
T-SNE at Works

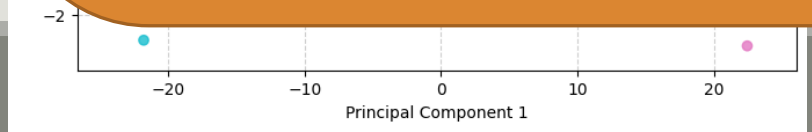
1) Set 4 centroids in a 5-dimension dataspace
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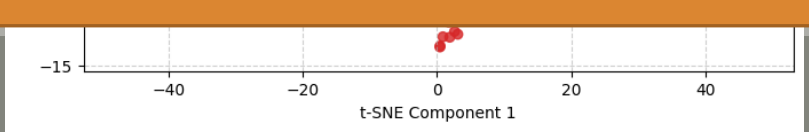
Global structure is lost, Local structure is preserved (e.g. we can distinguish blue from red cluster but the reds and blues are still near)



Adding **2 high variance random** features to the original dataset







When Does t-SNE Fail?

When you need a reusable mapping (out-of-sample data)

- t-SNE does not learn a function f(x)
- You can't take a new point and map it into the existing embedding, you'd rerun t-SNE on the whole dataset

When global structure matter

- st-SNE preserves local neighborhoods, not global geometry
- Distances between clusters are often meaningless
- Cluster sizes may be distorted

keep all information about the points

When you want quantitative analysis (not just visualization)

- The axes have no meaning
- Distances are not reliable metrics

When results must be stable and reproducible

(different run of t-SNE bring different results)

- t-SNE depends on random initialization
- Sensitive to hyperparameters (especially perplexity)

See [Eating_in_the_UK.ipynb]

Eating in the UK

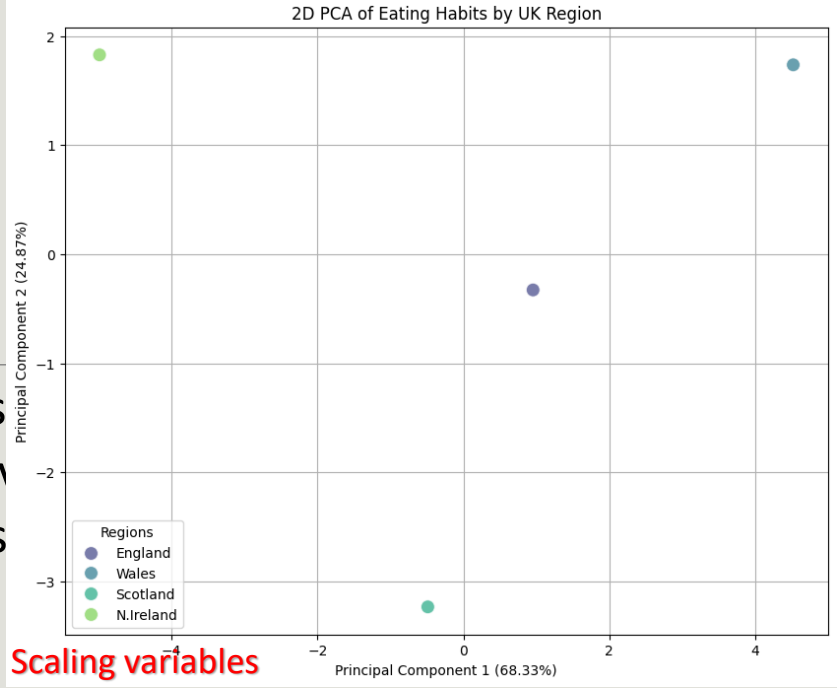
What if our data have way more than 3-dimensions? Like, **17** dimensions?! In the table is the average consumption of 17 types of food in grams per person per week for every country in the UK.
Political Compass goes beyond dimensionality a reduction problem since it “forces” political questions into the two axes

	England	N Ireland	Scotland	Wales
Alcoholic drinks	375	135	458	475
Beverages	57	47	53	73
Carcase meat	245	267	242	227
Cereals	1472	1494	1462	1582
Cheese	105	66	103	103
Confectionery	54	41	62	64
Fats and oils	193	209	184	235
Fish	147	93	122	160
Fresh fruit	1102	674	957	1137
Fresh potatoes	720	1033	566	874
Fresh Veg	253	143	171	265
Other meat	685	586	750	803
Other Veg	488	355	418	570
Processed potatoes	198	187	220	203
Processed Veg	360	334	337	365
Soft drinks	1374	1506	1572	1256
Sugars	156	139	147	175

Eating in the UK

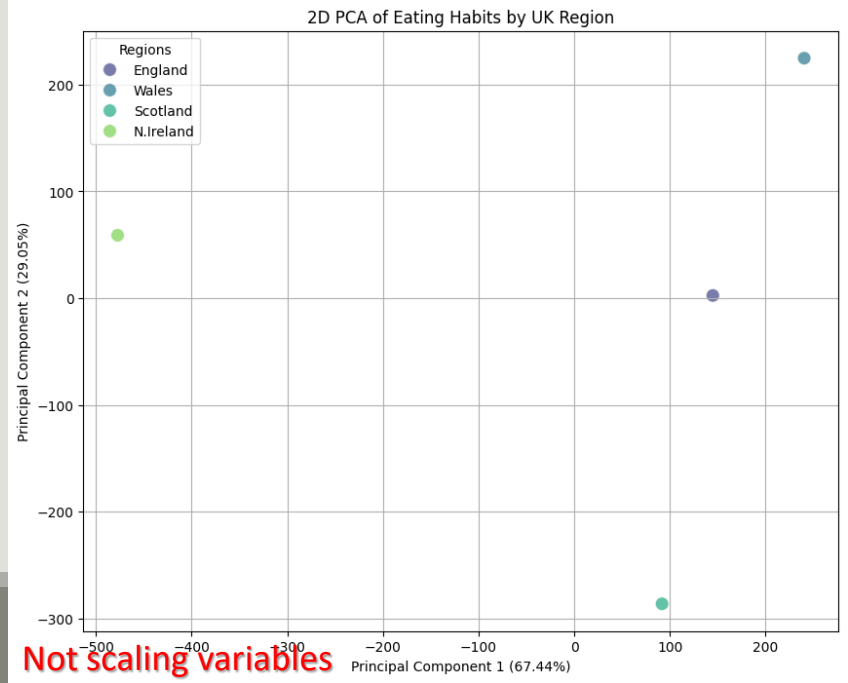
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Scaling variables

Explained variance: [0.68327871; 0.24871288] = 0.93



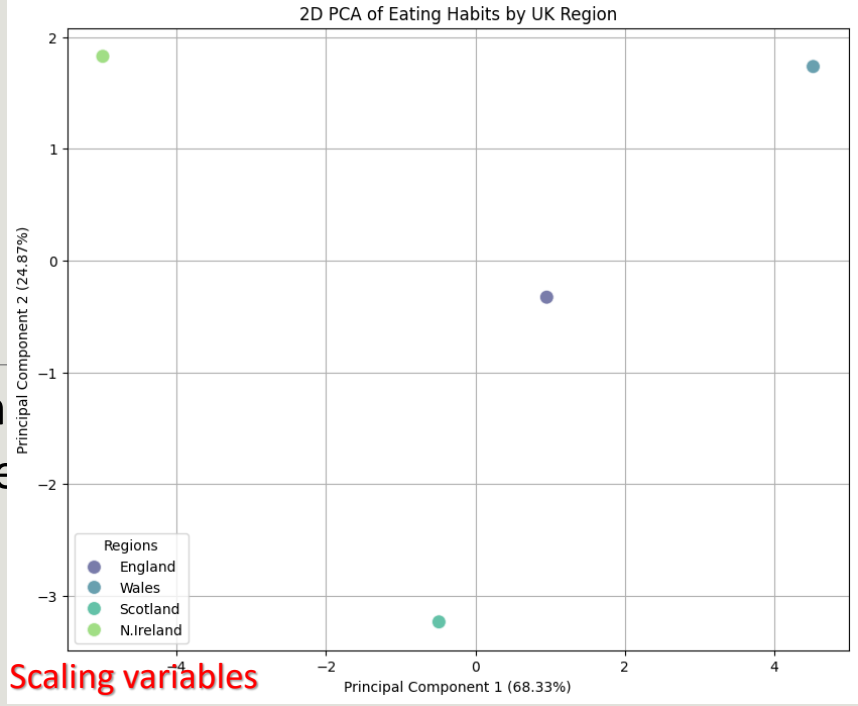
Not scaling variables

Eating in the UK

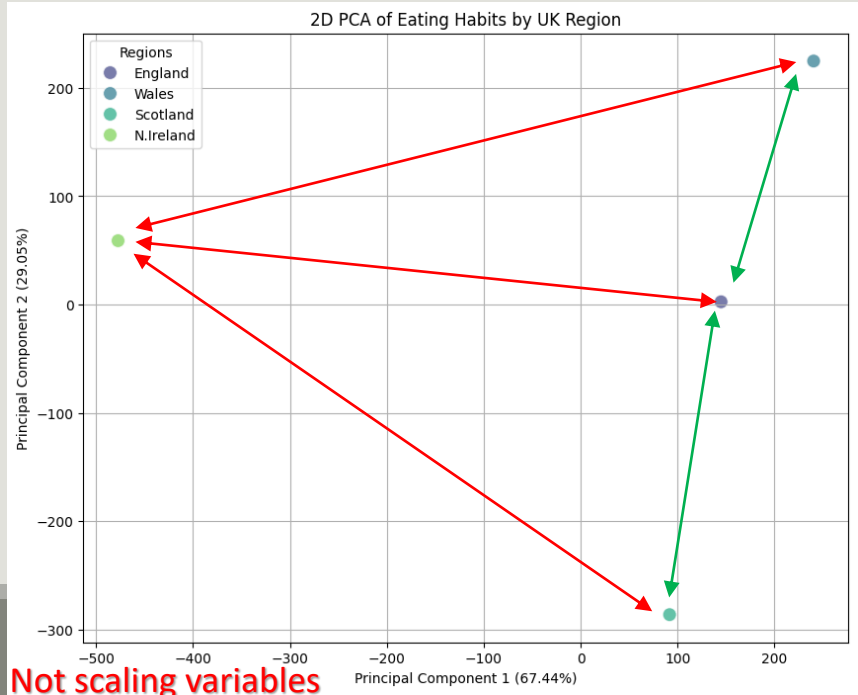
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the Northern Irish eat way more grams of fresh potatoes and way fewer of fresh fruits, cheese, fish and alcoholic drinks.



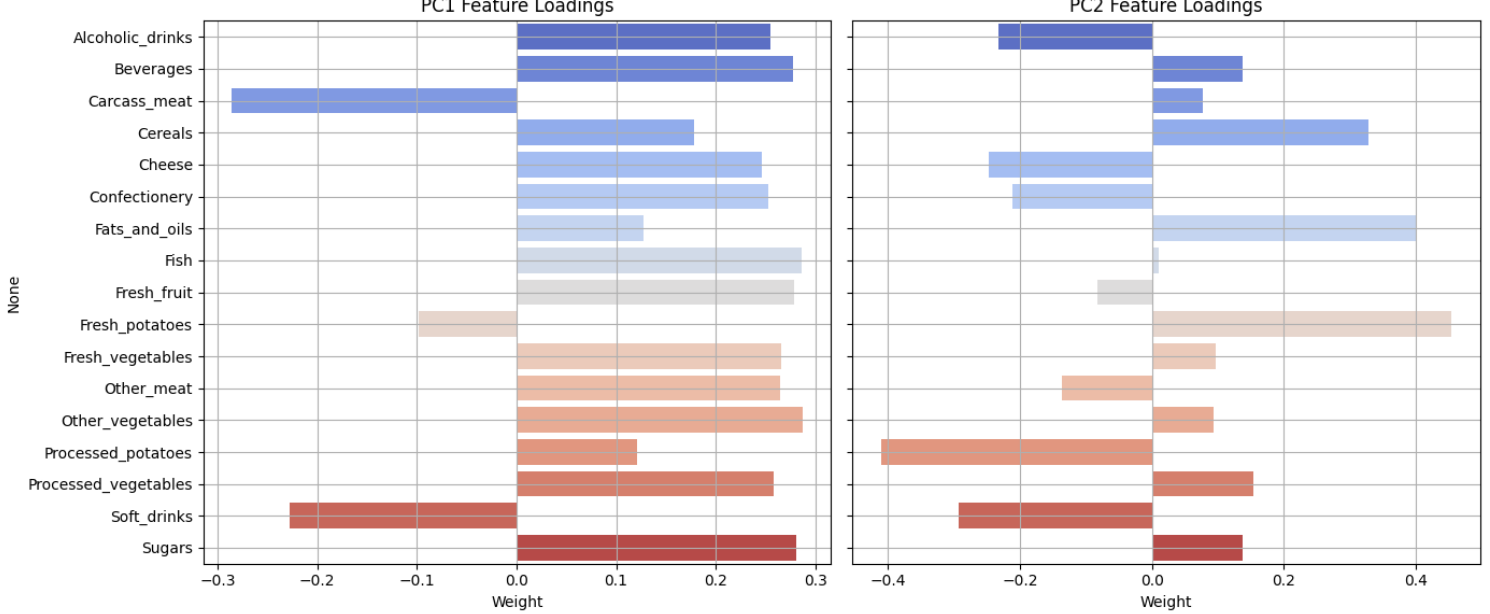
Scaling variables



Not scaling variables

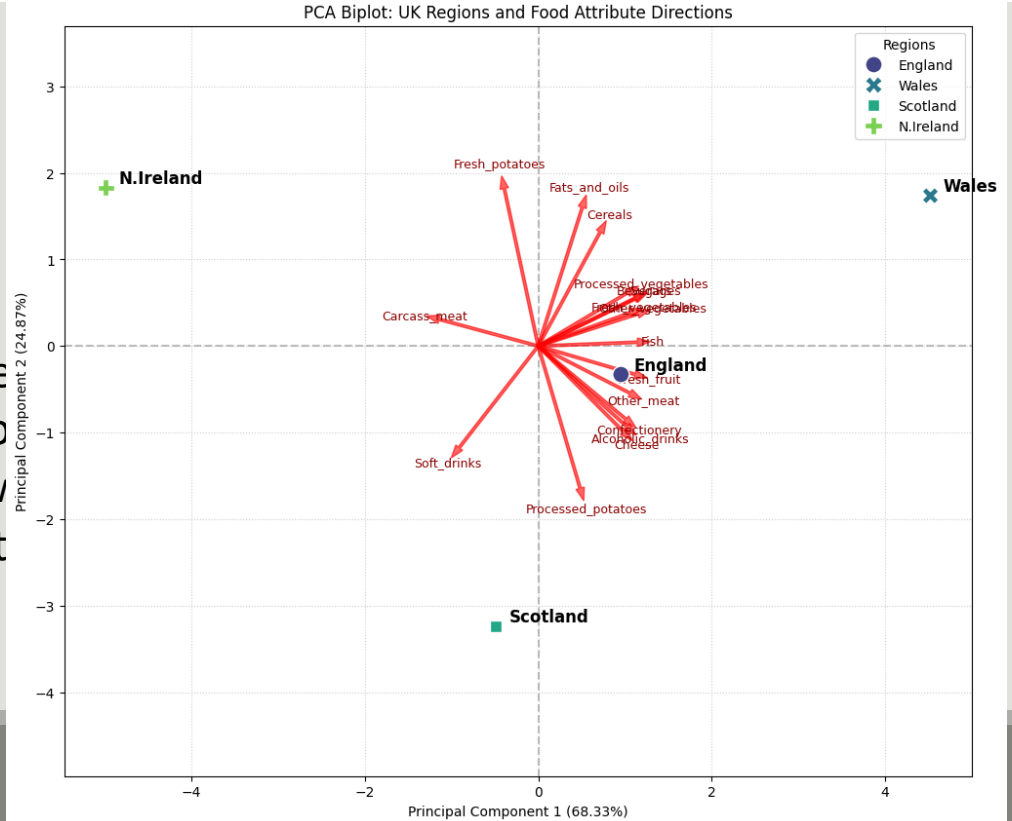
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The Political Compass Example

The **Political Compass** is a model used to represent political ideology in a two-dimensional space, rather than the traditional single left–right line. It is based on 62 propositions across economic + social axes. The two axes involved are:

- Economic axis (left–right)
- Social axis (authoritarian–libertarian)

Political Compass goes beyond dimensionality a reduction problem since it “forces” political questions into the two axes

Spatial Modeling of Politics

Let's create a Spatial Model of Politics Position

1. Define the questionnaire (10 dimensions)
   - These questions represent the 10-dimensional feature space
   - *Questions define the dataspace. If you change the questions everything will change*
2. Collect answers from citizen
   - Each individual is a data point in the 10-dimensional space
3. Define political party profiles
   - This encodes the ideological "position" of each party in the same feature space as individuals
   - *Parties are "human feature engineering" not data!*
4. Apply dimensionality reduction
   - Visualize global structure
   - Identify general directions of variation in the data
   - Analyze how individuals relate to political parties in the reduced space
   - *Latent axes must be evaluated they do not have a predefined meaning (e.g. Left vs Right)*

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To what extent do you agree with the following statements?

- *"Lower taxes are more important than expanding social welfare programs.*
- *"Immigration should be more strictly regulated and limited.*
- *"Environmental protection should be prioritized even if it slows economic growth.*
- *The state should play a larger role in regulating the economy than the free market*

Progressive
Conservative
Liberal
Populist

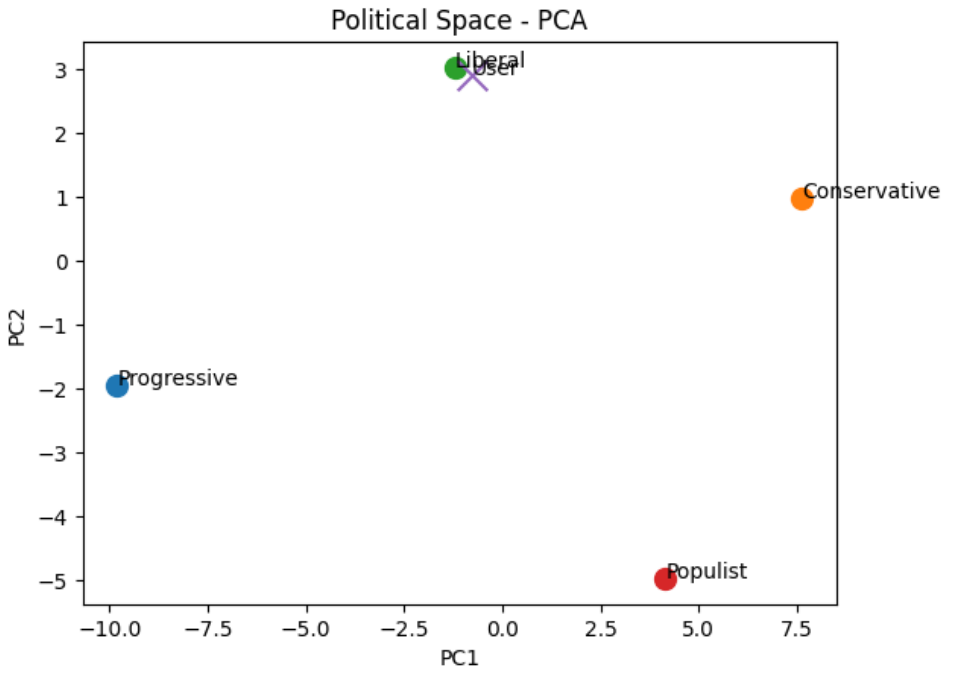
See Spatial_Modeling_of_Politics.ipynb

Spatial Modeling of Politics

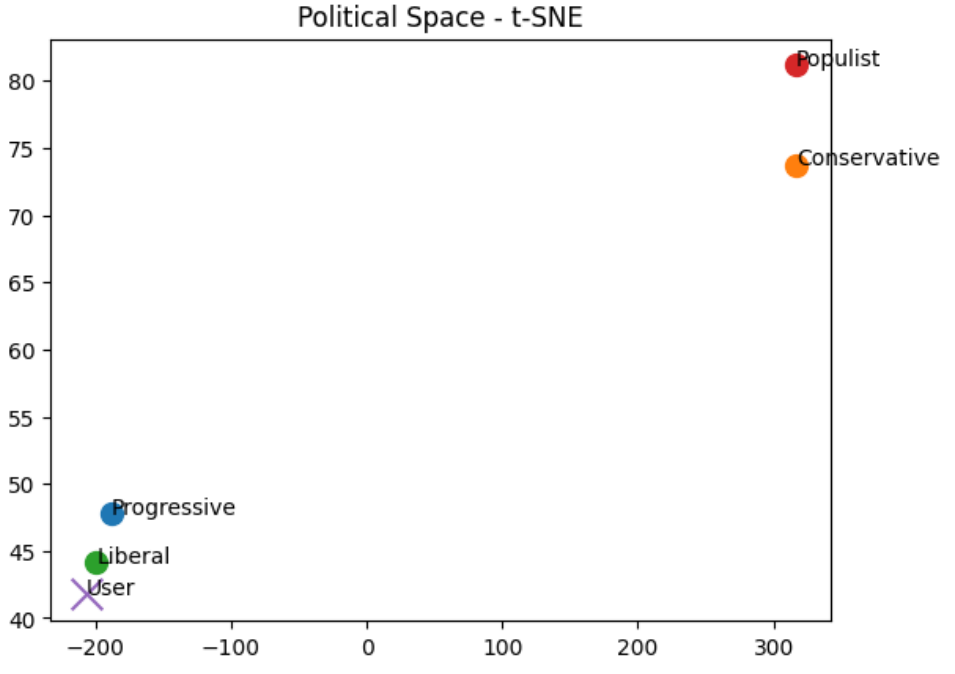
Parties vs User

```
"Progressive":  [2, 9, 8, 9, 9, 8, 3, 3, 8, 9],
"Conservative": [9, 3, 4, 3, 3, 4, 9, 9, 4, 3],
"Liberal":      [7, 5, 8, 6, 8, 5, 5, 8, 7, 5],
"Populist":     [8, 7, 2, 4, 4, 7, 9, 6, 3, 6],
"user":         [6, 6, 7, 4, 5, 3, 4, 4, 7, 4]
```

10D euclid. distance	**Progressive**	**Conservative**	**Liberal**	**Populist**	**User**
Progressive	0.00	17.69	10.30	14.35	10.95
Conservative	17.69	0.00	9.43	7.14	9.64
Liberal	10.30	9.43	0.00	10.10	6.16
Populist	14.35	7.14	10.10	0.00	9.80
User	10.95	9.64	6.16	9.80	0.00



Explained variance: [0.71571895 0.19383102]

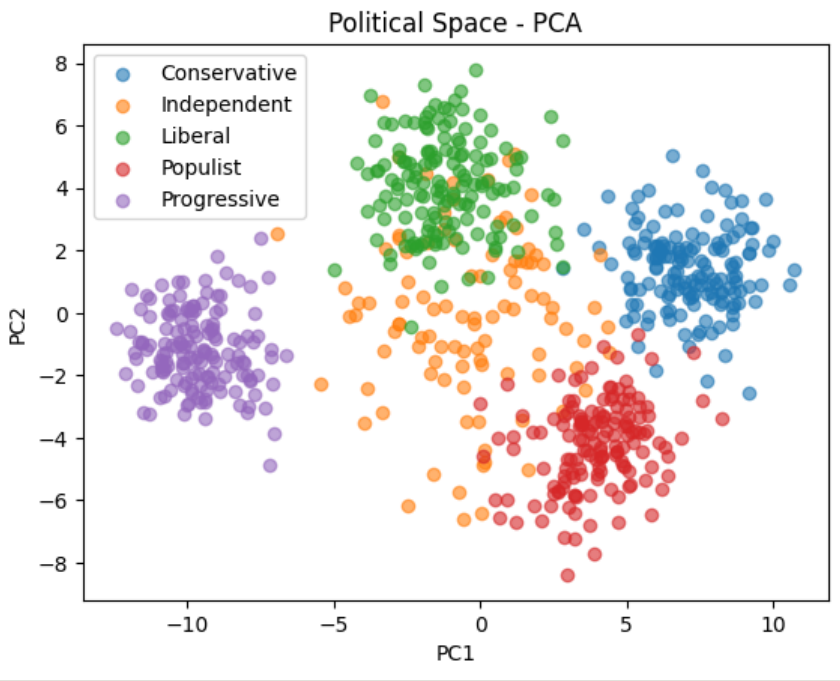


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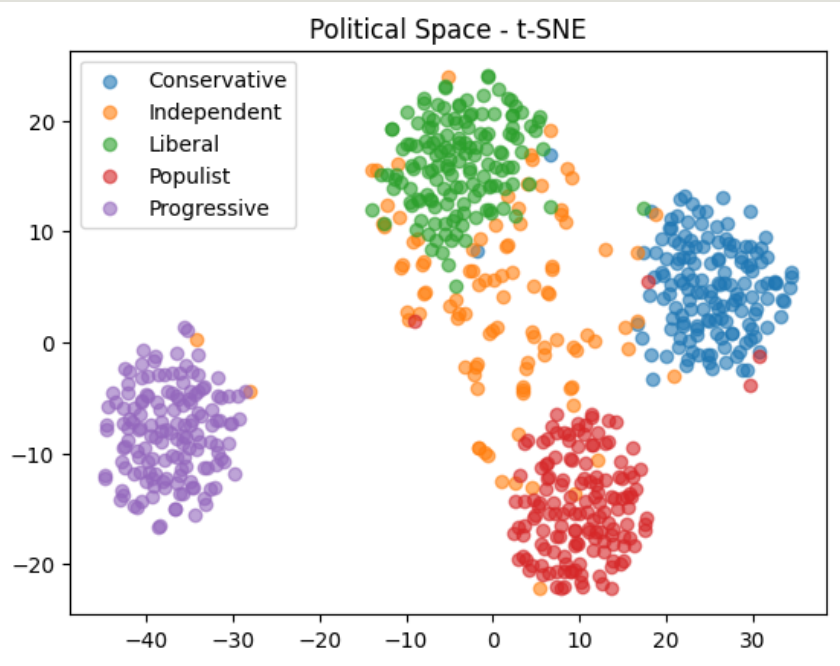
Spatial Modeling of Politics

Parties vs Citizen

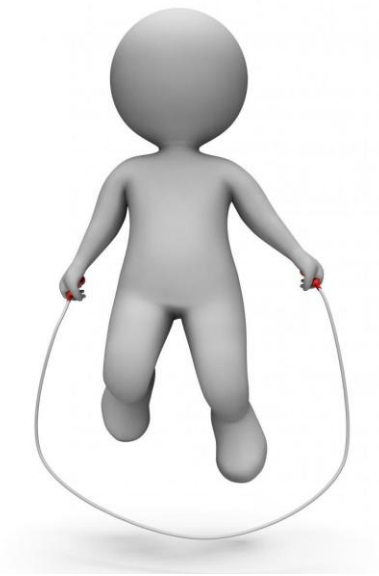
[8, 3, 6, 4, 3, 3, 6, 8, 4, 2],
[8, 8, 1, 2, 5, 9, 7, 8, 3, 8],
[4, 5, 4, 7, 9, 2, 4, 2, 7, 8],
[8, 6, 4, 3, 2, 8, 8, 5, 1, 5],
[7, 9, 3, 7, 6, 9, 7, 7, 3, 5],
[9, 8, 2, 6, 3, 8, 7, 5, 1, 6],
[2, 8, 6, 9, 8, 6, 2, 1, 10, 10],
[7, 7, 9, 4, 10, 5, 6, 10, 5, 5],
[4, 8, 8, 9, 8, 9, 3, 4, 7, 8],
[3, 10, 7, 8, 7, 5, 6, 1, 8, 9],
.....



Explained variance (PCA): [0.5298842 0.15313032]



The Titanic Dataset



[The kaggle dataset]

The sinking of the Titanic is one of the most infamous shipwrecks in history.

On April 15, 1912, during her maiden voyage, the widely considered “unsinkable” RMS Titanic sank after colliding with an iceberg. Unfortunately, there weren’t enough lifeboats for everyone on board, resulting in the death of 1502 out of 2224 passengers and crew.

While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.

